The Long Ascent

A Kerbal Space Program Modlist & Guide

Version 1.0 — July 2026

For Kerbal Space Program 1.12.5

Compatible with Breaking Ground & Making History

Prologue

We used to look up at the sky and wonder at our place in the cosmos. Now we look down and wonder at ours. KSP gives you a universe and asks: what will you do with it?

That question has no right answer. You might land on the Mun and call it a career. You might build a station, a colony, an empire spanning star systems. You might strand Jebediah on Eve and spend the rest of your save trying to bring him home. All of these are valid. All of them are the point.

This guide is not a checklist. It's a companion. It will teach you to fly, to land, to dock, to mine, to build, to travel between stars. But between the instructions, it will also remind you: stop. Look around. You are standing on a moon orbiting a gas giant in a simulated solar system you assembled yourself from a hundred mods written by strangers who love space as much as you do. That's not a loading screen. That's a miracle.

The mods in this list were chosen to serve a feeling — the awe of exploration, the solitude of deep space, the terror and beauty of the unknown. You'll hear it in the engine roar at liftoff. You'll see it in the accretion disk of a black hole. You'll feel it in the silence between stars.

The Long Ascent is not just a journey upward. It's a journey outward. Into the dark. Into the strange. Into the places Kerbals have never been.

Take your time. There's no hurry. The stars will wait.

"Do not go gentle into that good night."

[— Dylan Thomas]

Before You Begin

On Pacing

KSP is a slow game. This is not a bug. It is the entire point.

The films that inspired this guide — 'Interstellar', '2001: A Space Odyssey', and 'Project Hail Mary' — share a quality that most games lack: they let moments breathe. A docking sequence takes four minutes of screen time. An arrival at a new planet holds on the image. Silence is used as punctuation.

You should play KSP the same way.

- **Do not rush.** Time warp is a tool, not a lifestyle. During long burns, switch to IVA view and watch the stars crawl past the window. During interplanetary cruises, check on your kerbals. Read their mission logs if you keep them. Let the journey take the time it takes.
- **Look at things.** The mods in this list exist to make the Kerbol system beautiful. Scatterer paints the atmospheres. Parallax textures the ground. Volumetric Clouds turn the sky into sculpture. If you spend an entire mission staring at the navball, you are missing the point.
- **Earn your moments.** A Mun landing is not a checkbox. It is the culmination of learning orbital mechanics, building a rocket that doesn't explode, and navigating 400,000 kilometers of vacuum to touch another world. When you plant that flag, stop. Pan the camera. Look back at Kerbin, hanging in the sky. You did that.
- **Embrace the silence.** Deep space is quiet. Between the engine burns and the alarm bells, there are long stretches of nothing. Those stretches are where the awe lives. Don't fill them with distractions.

On Failure

You will crash. You will strand kerbals. You will spend three hours on a mission that ends in a fireball because you forgot one parachute. This is not a failure state. It is the learning process.

The guide does not expect you to execute every maneuver perfectly on the first try. It expects you to revert, redesign, re-launch, and try again. The difference between a beginner and an expert is not success rate. It is how quickly they say “well, that didn't work” and return to the VAB.

On the Waves

This modlist is structured in four cumulative waves. Each adds complexity, parts, and destinations. But more importantly, each adds a new kind of experience:

- **Wave 0:** The thrill of reaching orbit for the first time. The moment you realize you're no longer ground-bound.
- **Wave 0.5:** The precision of planning. The satisfaction of docking. The quiet pride of watching your space station orbit overhead.
- **Wave 1:** The scale of interplanetary travel. The first time you see Jool rise over Laythe's ocean. The terror of Eve ascent.
- **Wave 2:** The solitude of interstellar flight. The wonder of another star. The weight of keeping kerbals alive across decades.

Install one wave at a time. Play it until the experiences it offers feel familiar. Then install the next. The guide will be here when you're ready.

The guide assumes patience. The guide assumes curiosity. The guide assumes you want to go further than anyone has gone before.

[Let's begin.]

Contents

Installation Guide

Prerequisites

- **Kerbal Space Program 1.12.5** — installed via Steam, GOG, or direct download
- **CKAN** — download from https://github.com/KSP-CKAN/CKAN/releases
- **Breaking Ground** and **Making History** DLCs — recommended but not required

Step-by-Step Setup

Step 1 — Install CKAN

Download the latest CKAN release for your platform. Launch CKAN. On first run it will ask you to locate your KSP installation — poinFt it at your KSP 1.12.5 directory.

> **Steam users:** Your KSP directory is typically at: `C:\Program Files (x86)\Steam\steamapps\common\Kerbal Space Program` (Windows) or `~/.steam/steam/steamapps/common/Kerbal Space Program` (Linux). Use File → Manage KSP Instances to manage multiple versions.

Step 2 — Add Repository (if needed)

CKAN should already include the default repository. If you don't see mods appearing:

1. Settings → CKAN Settings → New Repository
2. Add: `https://github.com/KSP-CKAN/CKAN-meta/archive/main.tar.gz`
3. Click OK, then File → Refresh

Step 3 — Install The Long Ascent Waves

The modlist ships as four CKAN metapackage files — one per wave — plus a combined file. Each wave is cumulative: Wave 0.5 includes Wave 0; Wave 1 includes all previous; Wave 2 is the complete modlist. Install them in order and play through each wave's guide before adding the next.

Wave files

- **Wave 0 — First Steps:** `the-long-ascent-wave-0.ckan` (8 mods). UI and HUD overlays: KER, BetterBurnTime, bugfixes, dark theme. No gameplay changes. Install first and learn orbital flight.
- **Wave 0.5 — Building Blocks:** `the-long-ascent-wave-05.ckan` (23 mods, cumulative with Wave 0). Planning tools and editor enhancements: TWP, DPAI, Trajectories, KAC, SCANsat, VAB tools. No new parts or mechanics. Install once you can reach orbit consistently.
- **Wave 1 — Going Further:** `the-long-ascent-wave-1.ckan` (77 mods, cumulative with Waves 0–0.5). Graphics overhaul (Scatterer, AVP, Deferred, Parallax, Waterfall), parts expansion (Near Future suite, Restock+, historical and modern rocket packs), planet packs (OPM, Kcalbeloh, Grannus, QuackPack), and fuel system mods. Install once you've mastered Mun landings, docking, and space stations.
- **Wave 2 — The Long Ascent:** `the-long-ascent-wave-2.ckan` (7 mods, cumulative with all previous). Life support, colonization, interstellar propulsion (FFT, Blueshift). Each mod is optional. Install once interplanetary travel is routine.

For each wave:

1. File → Install from .ckan... → select the wave's `.ckan` file
2. CKAN will display a list of mods — including dependencies from previous waves if this is your first install
3. Review the changes. CKAN resolves all dependencies automatically.
4. Click Continue, then Apply. CKAN downloads and installs everything.

First-time setup: Start with Wave 0 only. Play through the Wave 0 guide until you can reach orbit without reverting. Then install Wave 0.5, play its guide, and so on. Each wave's guide assumes you've mastered the skills from the previous wave.

Alternatively, use `the-long-ascent.ckan` (the combined file) to install all waves at once. All mods appear as recommendations — review the list, uncheck anything you don't want, and apply. The combined file includes everything from Wave 0 through Wave 2.

Step 4 — Launch and Verify

Launch KSP. When the main menu loads, you should see the ModuleManager banner in the lower-right corner, confirming mods are active. Start a new save — do not load old saves without checking compatibility.

> **Recommended first save:** Career mode with default settings for Wave 0. For Waves 0.5 and beyond, consider increasing science rewards slightly (120–150%) since additional mechanics and the Community Tech Tree (Wave 1) demand more science than stock. Funds rewards can stay at 100% — contracts scale well with the expanded parts list.

CKAN Troubleshooting

CKAN can't find my KSP install

CKAN looks for `buildID64.txt` or `KSP.exe` / `KSP.x86_64` in the game directory. Use File → Manage KSP Instances → Add New and manually browse to the folder.

Mods fail to download

Some mods are hosted on services that may be temporarily down. CKAN will show an error as it progresses — note which mod failed and try again later.

Conflict warnings

CKAN may warn about mod conflicts. When installing via the `.ckan` metapackages, conflicts are handled. If you see unexpected conflicts, check the `conflicts` notes in the individual mod entries later in this guide.

[WAVE 0]

[First Steps]

[UI & Quality of Life — Learning to Fly]

Wave 0 — Mod List

About Wave 0

Wave 0 is for new players or veterans returning to stock KSP. It adds only the most essential HUD, QoL, and bugfix mods — nothing that changes gameplay, parts, or physics. By the end of this wave you will reach stable Kerbin orbit, the foundation for everything that follows.

Criterion	Wave 0	Wave 0.5	Wave 1	Wave 2
Gameplay changed?	No	No	Adds content	Adds complexity
New parts?	No	No	Yes (2000+)	Yes (colony + interstellar)
Learning curve	None	Tools & planning	Parts, planets, graphics	Life support, colonies, FTL
Resource chains?	No	No	Fuel types (CRP/cryo)	Full (LS, colony, reactor fuel)
Failure conditions?	No (revert only)	No (revert only)	Mission failure (no revert)	Permanent (dead kerbals)
Progression goal	Stable orbit	Mun, docking, stations	Interplanetary	Interstellar, self-sustaining
Save-breaking?	No	No	Unlikely	Likely (choose wave carefully)

> **Wave 0 Decision Rule:** Only UI/HUD overlays, readouts, and bugfixes. No docking aids, no VAB tools, no planning utilities. If a mod changes how you build or plan rather than what information you see, it belongs in Wave 0.5 or later.

Mod List

Kerbal Engineer Redux

Impact: UI

Displays critical flight data (delta-v, TWR, orbital info, biome, and more) in a customizable HUD. Essential for knowing your rocket's capabilities before you launch.

Version: Compatible with KSP 1.12.x.

Adds readout panels to VAB/SPH and flight view. No new parts, no gameplay changes. Replaces guesswork with numbers.

BetterTimeWarp Continued

Impact: UI

Version: Compatible with KSP 1.12.x.

Adds custom warp speed levels and fixes stock time warp behavior. Lets you fine-tune warp rates for long burns or interplanetary transfers.

Replaces the stock warp rate list with customizable levels. No gameplay changes — just more control over how fast you wait.

KSP Community Fixes

Dependencies: ModuleManager, Harmony2

Impact: UI

Version: Compatible with KSP 1.12.x. Requires Harmony2. Settings configurable via the in-game difficulty options menu.

Comprehensive bugfix and QoL patch collection. Fixes dozens of stock bugs (resource handling, physics quirks, UI glitches) and adds optional quality-of-life tweaks like better maneuver node handling, SAS improvements, and part-action-window enhancements.

Patches stock bugs and adds optional UI/QoL improvements. All fixes can be individually toggled via config. No new parts or mechanics — just makes the stock game work better.

Station Keeping

Dependencies: ModuleManager

Impact: UI

Version: Compatible with KSP 1.12.x. Without it, every timewarp session slowly degrades your orbits. Essential once you start building stations and relay networks in later waves.

Fixes orbit drift during timewarp. In stock KSP, a vessel's orbit subtly changes every time you enter and exit timewarp — this mod locks orbits precisely on rails during warp so your carefully-placed station or satellite constellation stays exactly where you left it.

Prevents orbital drift during timewarp transitions. No new parts or mechanics — purely a precision fix for the stock physics engine. Critical for maintaining station orbits and relay networks.

BetterBurnTime

Shows more accurate burn time and countdown information next to the navball during maneuvers. Displays time-to-burn, estimated burn duration, and

Dependencies: ModuleManager

Impact: UI

Version: Compatible with KSP 1.12.x. Does not conflict with KER — provides complementary burn-time data displayed directly on the navball.

a "time-to-impact" readout during landings or rendezvous approach.

Improves the burn-time indicator next to the navball with more accurate predictions and additional context. No parts, no gameplay changes — just better numbers.

Precise Maneuver

Impact: UI

Version: Compatible with KSP 1.12.x. Complements MechJeb's Maneuver Planner (Wave 1) — use both for full maneuver control.

Replaces and overhauls the stock maneuver node editor with a precision window. Edit maneuver components numerically (prograde, normal, radial), fine-tune burn timing, and increment/decrement with configurable step sizes. For players who want exact numbers rather than dragging gizmos.

Replaces the stock maneuver gizmo with a numerical editor window. No gameplay changes — the maneuver node system works identically, you just get precise numeric control instead of click-and-drag.

Toolbar Controller

Impact: UI

Version: Compatible with KSP 1.12.x. Required dependency for mods in later waves.

Modern maintained toolbar framework. Provides a unified toolbar button system used by many mods (KAC, Waypoint Manager, Trajectories). Required dependency for several Wave 0.5 mods.

Provides a framework for toolbar buttons. No visible impact on its own — other mods use it to add their buttons.

ZTheme

Impact: UI, Graphics

Version: Compatible with KSP 1.12.x. Installed via CKAN. No dependencies. Can be toggled or uninstalled at any time without affecting saves.

Dark UI theme for KSP's interface. Replaces the stock light-gray UI panels with a sleek dark theme across all scenes. Easier on the eyes during long play sessions.

Replaces UI textures and color schemes with a dark theme. No gameplay changes — purely cosmetic UI reskin. Applies globally to all KSP interface elements.

Wave 0 Mods — How to Use Them

Kerbal Engineer Redux (KER)

KER adds two things: a toolbar button in the VAB/SPH, and a toolbar button in flight. Both open customizable readout panels.

In the VAB: Click the KER button to open the build engineer panel. This shows your rocket's delta-v per stage, TWR, and burn time — all calculated live as you build. You'll see two delta-v columns: **atmospheric** and **vacuum**. Atmospheric is for the first 20 km of ascent (engines are less efficient in thick air). Vacuum is for everything above. For transfer stages and landers, always plan using the vacuum number.

The most important VAB readouts: **Delta-V**, **TWR**, and **Burn Time**. If TWR is below 1.0 at launch, you won't leave the pad. If delta-v to orbit is below 3,400 m/s, you're cutting it close.

In flight: Press the KER toolbar button (or the hotkey you set in settings) to open the flight HUD. The essential flight readouts for a beginner:

- **Orbit: Apoapsis / Periapsis** — watch these during ascent and after maneuvers
- **Vessel: Altitude (Surface)** — shows your true height above terrain, not sea level
- **Surface: Vertical Speed** — critical for landings. Keep this low and decreasing
- **Orbit: Time to Apoapsis / Periapsis** — timing your circularization burns
- **Vessel: Total Delta-V** — how much gas you have left

> KER's readouts are deeply customizable, but don't go overboard. Start with the five listed above. Add more as you learn what each number means. A cluttered HUD is harder to read than no HUD at all.

BetterTimeWarp Continued

BetterTimeWarp replaces the stock warp bar (the arrows next to the MET clock) with a customizable warp menu. Click the toolbar button (a clock icon) to open settings.

Key features:

- **Custom warp rates:** You can add intermediate speeds (8×, 15×, etc.) for more granular time compression.
- **Physics warp override:** Normally, physics warp (Alt+period) caps at 4×. Better-TimeWarp lets you push it higher.
- **Lossless physics warp:** At moderate rates (up to 10× physical), BetterTimeWarp runs extra physics steps to reduce wobble and explosion risk.

> Set a custom physics warp of 6× for long ion-engine burns. At 6×, most craft can handle the physics load without spontaneous disassembly. Above 10×, save first.

Precise Maneuver

Open Precise Maneuver by clicking its icon (a maneuver node with a gear) in the toolbar when you have a maneuver node selected. The window shows numerical values for prograde, normal, and radial components. Use the +/− buttons for fine adjustments, or type exact values. The "Snap" dropdown controls increment size — use 1.0 m/s for rough planning, 0.1 or 0.01 for fine-tuning intercepts.

Station Keeping

Station Keeping works automatically — install it and forget it. Your station and satellite orbits will remain stable across timewarp sessions. No configuration needed. If you notice a station's orbit still drifting, check that the mod is installed correctly (look for a "StationKeeping" folder in GameData).

BetterBurnTime

BetterBurnTime replaces the stock burn-time indicator next to the navball. When you create a maneuver node, it shows three numbers:

- **Time to burn:** Countdown until you should start burning. The indicator turns green when it's time.
- **Burn duration:** How long the burn will take at full throttle.
- **Time to impact:** Shows during landings — how many seconds until you hit the surface. Watch this on final descent.

Start your burn so that half the burn happens before the node and half after. BetterBurnTime handles the timing — when the countdown hits zero, throttle up. When the "Est. Burn" counter reaches zero, cut engines.

> During landings, BetterBurnTime's impact timer is more responsive than the altimeter alone. When it counts down to 3 seconds at high speed, you're about to lithobrake. Burn retrograde immediately.

Wave 0 Guide — First Steps

Before You Launch

Game Mode Choice

Start a **Science** or **Career** save on Normal difficulty. Sandbox is tempting but overwhelming — you unlock all parts at once without context. Science mode gates parts behind experiments, teaching you one set of parts at a time. Career adds funds and contracts, giving you goals and constraints that mirror real space programs.

> If you choose Career, use default settings. Don't touch the difficulty sliders yet — the stock balance is well-tested for new players.

Key Concepts

Before you build anything, let's get the terminology straight. These words will appear constantly throughout the guide — and the game.

Delta-V (Δv)

Delta-v is your rocket's **fuel budget**. It tells you how much your rocket can change its velocity, measured in meters per second (m/s). Think of it like the range on a car — you need enough to get where you're going.

Every maneuver costs delta-v:
- Getting from the launch pad to low Kerbin orbit: 3,400 m/s
- Transfer from Kerbin orbit to the Mun: 860 m/s
- Landing on the Mun from low orbit: 580 m/s
- Returning to Kerbin: 860 m/s

The sum of these is your mission's delta-v budget. If your rocket has less than the total required, you won't make it.

> There are two kinds of delta-v: **atmospheric** (inside a planet's atmosphere) and **vacuum** (in space). Atmospheric delta-v is always lower because engines are less efficient when fighting air pressure. Always check which number you're looking at — the VAB shows

> atmospheric by default. Switch to vacuum in the delta-v panel for transfer and landing calculations.

Apoapsis and Periapsis

Every orbit is an ellipse. The **apoapsis** (Ap) is the highest point of your orbit above the body. The **periapsis** (Pe) is the lowest point. You'll use these constantly in the Map view to plan burns.

- Burning prograde at apoapsis raises periapsis
- Burning prograde at periapsis raises apoapsis
- To circularize your orbit, burn at Ap or Pe until they're equal

Prograde and Retrograde

On the navball (the big blue ball at the bottom-center of your screen during flight):

- **Prograde** (green circle, no X) — your direction of travel. Burn prograde to speed up.
- **Retrograde** (green circle with an X) — opposite your direction of travel. Burn retrograde to slow down.

These are relative to what you're orbiting. In space, prograde is your orbital velocity vector. Near the ground, it's your surface velocity. Click the navball speed indicator to toggle between Orbit, Surface, and Target modes.

TWR — Thrust-to-Weight Ratio

TWR is the ratio of your engine's thrust to your rocket's weight. A TWR of 1.0 means you hover. Below 1.0, you don't leave the pad. For a comfortable launch, aim for a TWR between 1.3 and 2.0 at liftoff. Higher TWR means faster acceleration but can cause aerodynamic stress and overheating in the lower atmosphere.

> As you burn fuel, your rocket gets lighter, so TWR increases during flight. A rocket that starts at TWR 1.3 might be at TWR 4.0 by the time the first stage burns out. Watch your throttle — you may need to reduce it to avoid going too fast too low.

SAS — Stability Augmentation System

SAS is your autopilot. Press T to toggle it on. When active, it holds your orientation against external forces and can automatically point at specific directions (prograde, retrograde, radial, normal, target, etc.). Higher-level pilots and probe cores unlock more SAS modes.

The Navball

The navball shows your orientation in 3D space:

- **Blue half** — you're pointed at the sky
- **Brown half** — you're pointed at the ground
- The line where they meet is the horizon
- The center dot is your current heading
- Markers around the edge show cardinal directions: 0° = North, 90° = East, 180° = South, 270° = West

Most launches go **east** (90°) because Kerbin's rotation gives you a free 175 m/s of velocity in that direction.

The Kerbal Space Center

Click through the buildings to understand what each one does:

- **Vehicle Assembly Building (VAB)** — build rockets (vertical launch)
- **Spaceplane Hangar (SPH)** — build planes (horizontal takeoff)
- **Tracking Station** — view and control all active flights
- **Mission Control** — accept and review contracts
- **Administration Building** — strategies (ignore for now)
- **Research and Development** — unlock parts with science points
- **Astronaut Complex** — hire and manage kerbonauts
- **Launch Pad / Runway** — where craft go up

Building Your First Rocket

The Basics of Rocket Design

Open the VAB. You'll see the parts list on the left and the assembly area in the center. Every rocket needs:

1. **Command Pod** — where the kerbal sits. Start with the Mk1 Command Pod.
2. **Fuel Tank** — holds liquid fuel and oxidizer. The FL-T series is standard.
3. **Engine** — burns fuel to produce thrust. The LV-T45 "Swivel" is a good first engine (it gimbals — steerable thrust).
4. **Parachute** — for landing safely. Mk16 parachute on top of the pod.
5. **Decoupler** — separates stages. Place between the pod and the tank if you want the pod to return alone, or at the bottom to eject the entire stage.

> **The Golden Rule of Rocket Design:** Heavy stuff goes at the top, engines at the bottom. Your center of mass should be above your center of thrust. If the rocket flips during ascent, you have a stability problem — add fins at the bottom.

Staging

Staging is the sequence in which parts fire. The staging stack is on the right side of the VAB. The bottom-most stage fires first. A basic staging sequence:

- Stage 0 (top): Parachute deploys
- Stage 1: Decoupler fires, pod separates from tank
- Stage 2 (bottom): Engine ignites at launch

Drag parts in the staging list to reorder them. When in flight, press Space to activate the next stage.

The Science Jr. and Goo

On your first rocket, add a **SC-9001 Science Jr.** (materials bay) and a **Mystery Goo Containment Unit** attached radially to the fuselage. These generate science when activated in different situations (launch pad, low atmosphere, high atmosphere, space). Press the green clipboard icon in flight to review science opportunities.

Reaching Orbit

The Gravity Turn

Building a rocket that reaches space is easy. Reaching **orbit** requires speed — about 2,300 m/s sideways. The most efficient ascent profile is the gravity turn:

1. Launch vertically until you reach 100 m/s or 1,000 m altitude
2. Tilt eastward (toward the 90-degree heading mark on the navball) by about 5–10 degrees
3. Follow the prograde marker (the green circle without an X on the navball) — it will naturally drift toward the horizon
4. By 10,000 m, you should be at roughly 45 degrees pitch
5. By 30,000 m, near-horizontal
6. Switch to Map view (M) and watch your apoapsis. Cut the engine when apoapsis reaches 80,000 m (above 70 km = space)
7. Coast to apoapsis, then burn prograde to circularize

> If your rocket flips during the gravity turn: add fins at the bottom, make the rocket taller rather than wider, and keep your speed below 300 m/s while still in the thick lower atmosphere (below 10 km).

> **Look down.**

> You are in orbit. Not “almost in orbit.” Not “in space but falling back.” You are circling an entire planet at 2,300 meters per second, and it is not pulling you down. The blue arc of Kerbin fills the window. The stars are steady. You did this with math, engineering, and a rocket you built yourself.
>
> This is not a loading screen. This is a planet. And you are flying around it.

Where to Go from Here

At this point you can reach a stable Kerbin orbit. You understand the gravity turn, how to read the navball, and how to use KER’s readouts to plan your burns. Congratulations — you’re no longer ground-bound.

From here, you have options:

- **Practice:** Launch to different orbit altitudes and inclinations. Try a polar orbit (heading north instead of east).
- **Experiment:** Build progressively larger rockets. See how much payload you can lift to orbit.
- **Progress:** When you can consistently reach orbit without reverting flights, install Wave 0.5. It adds advanced planning tools and teaches you to go further — Mun landings, docking, and space stations.

Every landing is a controlled crash.

When BetterBurnTime's impact timer counts down and your engine bell kisses the surface, you are completing a journey that began on the launchpad. Fuel tanks emptied, stages discarded, parachutes deployed — every gram of your rocket was accounted for in the math that brought you here. The kerbal in the pod just traveled farther than most humans in history. Take a moment before you hit "Recover Vessel."

Player Challenges — Wave 0

These are optional goals to test your orbital skills. No mods beyond Wave 0 required.

- **Single-Stage to Orbit:** Reach orbit without decoupling anything. The entire rocket goes to space together.
- **SRB Only:** Orbit using only solid rocket boosters — no liquid engines. Throttle management is... creative.
- **Minimalist:** Orbit with a rocket under 10 parts. Every part must earn its place.
- **High Orbit:** Achieve a circular orbit at 500 km altitude (stock contracts consider this "high orbit" — science bonus).
- **Polar Orbit:** Launch north from KSC into a 90° inclination orbit. Harder than east because you get no rotation assist.
- **Precision Landing:** De-orbit and splash down within 5 km of the Kerbal Space Center. Use your trajectory prediction skill.

[WAVE 0.5]

[Building Blocks]

[Advanced Tools, Planning & Editor Enhancements]

Wave 0.5 — Mod List

About Wave 0.5

Wave 0.5 adds advanced planning tools, editor enhancements, and visual indicators. These mods go beyond Wave 0's pure UI/HUD — they change how you interact with the VAB/SPH, plan missions, and manage your save. By the end of this wave you will land on the Mun, dock spacecraft in orbit, and assemble your first space station. You must be able to reach orbit consistently (Wave 0) before installing these.

Criterion	Wave 0	Wave 0.5	Wave 1	Wave 2
Gameplay changed?	No	No	Adds content	Adds complexity
New parts?	No	No	Yes (2000+)	Yes (colony + interstellar)
Learning curve	None	Tools & planning	Parts, planets, graphics	Life support, colonies, FTL
Resource chains?	No	No	Fuel types (CRP/cryo)	Full (LS, colony, reactor fuel)
Failure conditions?	No (revert only)	No (revert only)	Mission failure (no revert)	Permanent (dead kerbals)
Progression goal	Stable orbit	Mun, docking, stations	Interplanetary	Interstellar, self-sustaining
Save-breaking?	No	No	Unlikely	Likely (choose wave carefully)

Wave 0.5 Decision Rule: These mods add no new parts or gameplay mechanics, but they change editor workflows, introduce mission planning, and affect how you build. If you're still learning the stock VAB and mission flow, stay on Wave 0 for now.

Mod List

Transfer & Trajectory

Transfer Window Planner

Impact: UI

Version: Compatible with KSP 1.12.x.

A porkchop-plot calculator for interplanetary transfers. Shows the most efficient departure date and delta-v required to reach any planet or moon. Generates ejection angles and suggested burn parameters for precision transfers.

Adds a planning tool accessible from the VAB and Tracking Station. No parts, no gameplay changes — purely informational. Shows delta-v over time as a porkchop plot with color-coded efficiency.

Trajectories

Dependencies: ModuleManager, ToolbarController

Impact: UI

Version: Compatible with KSP 1.12.x. Accuracy depends on vessel orientation during atmospheric flight. Set your attitude then read the prediction.

Predicts your vessel's atmospheric trajectory accounting for drag and planetary rotation. Shows the actual landing site on the planet's surface in Map view — essential for precision landings at the KSC runway, targeted reentries, or Falcon 9-style booster recovery.

Overlays a predicted atmospheric trajectory on the Map view. Shows where you'll actually land, accounting for aerodynamics and rotation. Purely predictive — does not change flight physics.

Docking Aids

Docking Port Alignment Indicator

Dependencies: ModuleManager

Impact: UI

Version: Compatible with KSP 1.12.x. Wave 2 suggests switching to Community Navball DAI once docking becomes second nature.

Adds a dedicated docking alignment window showing relative position, orientation, and alignment of your vessel to the target docking port. Displays a crosshair-style indicator that makes precision docking intuitive. Recommended while learning docking.

Adds a popup window during docking with alignment crosshairs, distance, and relative angle readouts. No parts — the indicator appears when you target a docking port and select "Control from Here" on your own port.

⚠ Conflicts: Community Navball Docking Alignment Indicator provides similar information on the navball instead of in a separate window. Choose the interface style you prefer — you do not need both. DPAI is the recommended default for this modlist.

Science & Mission Tracking

[x] Science! Continued

Impact: UI

Version: Compatible with KSP 1.12.x. Essential for Career and Science mode — prevents the tedious process of manually tracking which experiments you've run where.

Science experiment checklist and tracker. Shows which experiments you've completed and which are still available per biome, situation, and celestial body. Eliminates the guesswork of "have I done a crew report in low Mun orbit?" — the checklist tells you exactly what science remains.

Adds a toolbar window listing all available science experiments with completion status. No new parts or gameplay changes — purely tracks what you've already done. Filterable by body, biome, and experiment type.

Mission Planning & Management

Kerbal Alarm Clock

Dependencies: ToolbarController

Impact: UI

Version: Compatible with KSP 1.12.x. Optional — the stock alarm clock (added in KSP 1.12) handles basic use cases. Use KAC if you manage 5+ concurrent missions.

Advanced alarm system for managing multiple concurrent missions. Set alarms for maneuver nodes, transfer windows, SOI changes, and periapsis/apoapsis crossings. Automatically pauses timewarp or kills warp before the event. More feature-rich than the stock alarm clock.

Adds an alarm clock interface accessible via toolbar. Alarms can trigger at specific times, orbit events, or transfer windows. No gameplay changes — purely a mission management tool.

Waypoint Manager

Dependencies: ToolbarController

Impact: UI

Version: Compatible with KSP 1.12.x. Waypoints persist across flights in the same save. Essential for precision surface operations and base-building in later waves.

Displays custom waypoints in the flight view and on the navball for navigation. Create waypoints at specific coordinates, at your current position, or at nearby vessels. Useful for marking landing sites, biome boundaries, or base locations during flight.

Adds in-flight waypoint markers visible in the world and on the navball. Waypoints are persistent and can be created/edited in flight. No parts or gameplay changes.

Tracking Station Evolved

Impact: UI

Version: Compatible with KSP 1.12.x. Essential once you have 20+ active flights. Particularly useful for relay networks and multi-mission campaigns.

Overhauls the Tracking Station interface with sortable, filterable vessel lists, customizable columns, and bulk vessel management. When you have dozens of flights, debris, and stations across the Kerbol system, the stock tracking station becomes unwieldy — this fixes that.

Replaces the stock Tracking Station vessel list with a sortable, filterable table. Add/remove columns for orbit info, crew, resources, and vessel type. Bulk-terminate debris, rename vessels, and sort by any parameter.

CapCom – Mission Control On The Go

Dependencies: ToolbarController

Impact: UI

Version: Compatible with KSP 1.12.x. Manual install from SpaceDock — not on CKAN. Requires ToolbarController.

Review and accept contracts without returning to Mission Control. Access the contract list from any scene via a toolbar button — accept new contracts, review active ones, and check deadlines while in flight or the tracking station.

Adds a toolbar button to access the contract list from any scene. No gameplay changes — contracts still function identically, you just don't need Mission Control for management.

Contract Configurator

Impact: UI

Version: Compatible with KSP 1.12.x. Framework only — no effect without contract packs. Install now so it's ready when you add career-expanding packs.

Framework for custom contract packs. Does nothing on its own — other mods use it to add new contract types (exploration, tourism, base-building, etc.) to Mission Control. Required if you want to expand KSP's career mode contract variety beyond the stock offerings.

Provides the contract infrastructure that custom contract packs build on. No gameplay changes by itself — install alongside contract packs to see new contracts. Framework-only mod.

VAB/SPH Tools

Conformal Decals

Apply customized flags and decals directly onto part surfaces. Decals conform to curved surfaces and

Impact: UI, Graphics

Version: Compatible with KSP 1.12.x. Works on any part surface. Decals are purely visual — no effect on vessel performance.

can be scaled, rotated, and positioned anywhere on your craft. Purely cosmetic — the Kerbal equivalent of nose art and mission patches on real spacecraft.

Adds decal parts that project your chosen flag onto any surface. Decals are physicsless — no mass, no drag, no gameplay impact. Complements TURD (recoloring) and SimpleRepaint (re-shading) for complete visual customization.

Buoyancy Adjuster

Dependencies: ModuleManager

Impact: UI

Version: Compatible with KSP 1.12.x. Makes submarine and underwater base construction feasible without part mass manipulation.

Adjust the buoyancy of any part to make submersibles, submarines, and floating bases practical. Stock KSP has no built-in buoyancy control — this mod lets you tune whether a part floats, sinks, or hovers at neutral buoyancy. Essential for underwater exploration.

Adds a buoyancy slider to part right-click menus. Positive values = float higher. Negative = sink. Zero = neutral. No new parts — every existing part gains configurable buoyancy.

Speed Unit Annex

Impact: UI

Version: Compatible with KSP 1.12.x. Particularly useful for aircraft builders — knowing you're at Mach 2.2 is more intuitive than 660 m/s.

Changes the navball speed readout from m/s to Mach number, knots, or km/h. Displays Mach number at the top of the navball when in atmosphere — essential for supersonic flight management and realistic aircraft operations.

Adds unit conversion to the navball speed indicator. No gameplay changes — purely changes how speed is displayed. Configure your preferred unit in the mod's settings. Mach is most useful for atmospheric flight; knots for maritime.

KerbNote Lite

Impact: UI

Version: Compatible with KSP 1.12.x. Terrain warnings are most useful for night landings,

Terrain warning system and in-flight notepad. Audible and visual terrain proximity warnings prevent CFIT (controlled flight into terrain) during landing and low-altitude flight. Also adds a persistent notepad for mission notes.

IVA-only flights, and landing on bodies with poor visibility.

Adds terrain proximity alerts with configurable warning thresholds. The “pull up” warning triggers when your descent rate and terrain proximity indicate imminent impact. Notepad feature persists across scene changes for mission tracking.

KerbVision IR

Impact: UI

Version: Compatible with KSP 1.12.x. Particularly useful for outer-planet missions (OPM, Jool) where solar illumination is extremely weak. Toggle on/off as needed — no permanent HUD change.

Night vision and infrared camera overlay for dark-side missions. Toggleable IR view illuminates terrain and vessels in complete darkness — essential for landing on a planet’s night side, docking in shadow, or navigating unlit terrain.

Adds a night-vision camera overlay accessible via toolbar. The IR view shows terrain and vessels in grayscale thermal-style rendering. No gameplay changes — purely a visual aid for low-light conditions.

The Janitor’s Closet

Dependencies: ModuleManager

Impact: UI

Version: Compatible with KSP 1.12.x. Particularly useful after installing large parts packs in Waves 1 and 2.

Parts management tool for the VAB/SPH. Filter, sort, and hide parts by mod, category, or custom rules. When you have dozens of parts packs installed, this keeps the editor part list manageable by letting you hide parts you never use.

Adds a part-filtering interface in the VAB/SPH. Can permanently hide parts from the editor list or temporarily filter by mod. No parts removed from the game — just hidden from the editor menu.

RCS Build Aid

Dependencies: ModuleManager

Impact: UI

Version: Compatible with KSP 1.12.x. Essential for designing balanced RCS systems, VTOL craft, and precision landers. Use the dry-CoM marker to ensure stability throughout all fuel states.

Shows your vessel’s center of mass with dry tanks (fuel drained) and visualizes RCS thrust vectors in the VAB/SPH. Displays torque caused by misaligned RCS — critical for building balanced craft that translate cleanly without unwanted rotation during docking.

Adds CoM visualization and RCS thrust-vector display in the editor. Shows how your CoM shifts as fuel drains and whether RCS placement will cause rotation. Pure build aid — no flight-time changes.

VAB Organizer

Impact: UI

Version: Compatible with KSP 1.12.x. Pairs with Janitor's Closet for complete editor part management. Most effective after installing large parts packs in Waves 1 and 2.

Organizes the VAB/SPH part list into collapsible, sortable categories far beyond the stock tabs. Group parts by mod, function, size, or custom rules. Complementary to Janitor's Closet — VAB Organizer reorganizes parts; Janitor's Closet hides them.

Reorganizes the editor part list with custom categories and sorting. Mod configs available for popular parts packs to auto-sort their parts into logical groups. Use alongside Janitor's Closet for full editor organization.

Editor Extensions Redux

Impact: UI

Version: Compatible with KSP 1.12.x. Higher symmetry counts useful for ring stations, engine clusters, and RCS placement. Angle snap customization enables precise truss and fairing angles.

Advanced build tools for the VAB/SPH: higher symmetry modes (up to 50×), angle snap customization, vertical/horizontal snapping, and fine-offset controls. Unlocks build precision that the stock editor's 8× symmetry and coarse angle snap can't achieve.

Extends the stock editor with additional symmetry modes, angle options, and alignment tools. No new parts — just more precise ways to place them. Enables builds that are impractical or impossible with stock editor constraints.

Hangar Extender

Impact: UI

Version: Compatible with KSP 1.12.x. Particularly useful in Wave 1 when building large historical launchers and space stations.

Expands the VAB and SPH build boundaries so you can construct rockets and planes larger than the stock editor allows. Essential for large historical rockets, interplanetary motherships, and massive space stations assembled in the editor.

Removes the editor build-area size limits. No new parts or mechanics — you can simply build larger. Extends build area in all directions — horizontal for SPH, vertical for VAB.

Water Launch Sites

Dependencies: KerbalKonstructs

Impact: UI

Version: Compatible with KSP 1.12.x. Manual install from fo-

Adds water-based launch sites to Kerbin. Launch seaplanes, boats, and amphibious craft directly from the ocean. Adds several water launch locations around Kerbin for varied maritime and seaplane operations.

Adds water launch sites via KerbalKonstructs. Purely additional launch location options for seaplane and maritime missions. Requires KerbalKonstructs.

rum thread — not on CKAN. Requires KerbalKonstructs (available via CKAN suggests in Wave 1).

Tech Tree

Hide Empty Tech Tree Nodes

Impact: UI

Version: Compatible with KSP 1.12.x. Most useful when paired with Community Tech Tree (Wave 1). Does nothing noticeable with the stock tech tree since all stock nodes are populated.

Removes empty tech tree nodes from the R&D Center view. When you install a modified tech tree like Community Tech Tree, not all nodes will be populated by your mods — this hides the empty ones for a cleaner interface.

Hides technology nodes in the R&D Center that have no parts available to unlock. Purely cosmetic — does not change the tech tree structure or part assignments.

Visual Indicators

IndicatorLights

Dependencies: ModuleManager

Impact: UI, Graphics

Version: Compatible with KSP 1.12.x. IndicatorLights Community Extensions adds indicators for mod parts (see below).

Adds functional LED indicators to stock parts. Batteries show charge level with colored lights, probe cores show signal status, docking ports show alignment, and more. Parts that were once opaque black boxes now visually communicate their state.

Adds colored light indicators to stock parts that reflect actual part state (charge, signal, control, etc.). Purely visual — part functionality is unchanged. The indicators provide at-a-glance status without opening right-click menus.

IndicatorLights Community Extensions

Dependencies: IndicatorLights

Impact: UI, Graphics

Extends IndicatorLights with LED indicators for popular mod parts including ReStock, Near Future Technologies, and others. Ensures your entire part catalog benefits from visual status indicators.

1

Version: Compatible with KSP 1.12.x. Requires IndicatorLights. Only useful if you install parts mods (Wave 1+).

Adds IndicatorLights compatibility patches for mod parts. No new functionality — just extends the visual indicator system to more parts. Requires IndicatorLights.

Wave 0.5 Mods — How to Use Them

Transfer Window Planner

TWP shows you optimal departure windows via porkchop plots — color-coded charts of delta-v cost over time. Pick the most efficient window and TWP provides the ejection angle and burn parameters. You'll use it constantly once you go interplanetary in Wave 1.

Docking Alignment

Install both DPAI and Navball DAI. Use DPAI's dedicated window while learning — the crosshair makes alignment intuitive. Once docking becomes automatic (usually after 10–15 successful docks), try switching to Navball DAI. The navball-integrated approach is faster once you can interpret the markers without thinking. Wave 2 suggests dropping DPAI entirely.

[x] Science!

Open the [x] Science! window from the toolbar in flight or at the Space Center. The main view shows every science situation (biome × experiment type) for your current body. Green checkmark = completed. Empty box = available, go get it. Red = impossible here. Filter by body to plan science-gathering missions efficiently. Use this before every landing to ensure you're not revisiting biomes you've already exhausted.

Kerbal Alarm Clock

KAC is most useful when you have 3+ concurrent missions. Create alarms for: maneuver nodes (reminds you 1 minute before burn), SOI changes (entering Duna's sphere of influence), transfer windows (don't miss the Duna window), and periapsis/apoapsis crossings (circularization points). The "Warp to Next" button handles timewarp automatically. KAC's advantage over the stock alarm clock is the ability to set alarms for events that haven't happened yet (future transfer windows, upcoming SOI changes).

VAB/SPH Tools

Janitor's Closet vs VAB Organizer: They solve different problems. Janitor's Closet **hides** parts you never use (e.g., hide all airplane parts when building rockets). VAB Organizer **reorganizes** the part list into better categories. Use both: Janitor's Closet to declutter, VAB Organizer to sort what's left.

RCS Build Aid: In the VAB, enable the CoM marker (bottom-left toolbar). RCS Build Aid adds a second "dry" CoM marker showing where the center of mass will be when tanks are empty. If CoM shifts dramatically as fuel drains, your RCS becomes unbalanced — add or reposition thrusters so they're equally distant from both the full and dry CoM.

Editor Extensions: The most impactful features are the extra symmetry modes (useful for engine clusters on large rockets) and angle snap customization. For trusses and custom fairings, 5° or 15° snap is often better than stock's 15°-only option. Vertical/horizontal snapping helps align radially-attached parts precisely.

Hangar Extender: Install before building large historical rockets (Saturn V, SLS) or interplanetary motherships. The extended boundaries are automatic — no configuration needed. If your craft is already hitting the VAB ceiling, save it as-is before installing, then reload and continue building.

Trajectories

Trajectories predicts where your craft will land — accounting for atmospheric drag and planetary rotation. Toggle it on from the toolbar (the icon looks like a trajectory arc).

- **Red X:** Predicted impact point. This is where you'll hit the ground if you do nothing.
- **Blue line:** Your trajectory through the atmosphere, showing how drag bends your path.
- **Body-fixed mode (default):** The X stays locked to the rotating planet surface. Use this for landings.
- **Inertial mode:** The X shows where you'd land if the planet stopped rotating. Use this for orbital planning only.

> When aerobraking at Duna or Jool, set Trajectories to body-fixed mode. Adjust your periapsis until the predicted exit trajectory has the apoapsis you want. A 15 km Duna periapsis captures you; a 25 km one just bends your path.

SCANsat

SCANsat adds persistent planetary maps. Open the map viewer from the SCANsat toolbar button (a radar dish icon).

- **Altimetry map:** Height above sea level. Use this to find flat landing zones and mountain peaks.

- **Biome map:** Color-coded biomes. Essential for science farming — plan your lander's route to visit multiple biomes per trip.
- **Slope map:** Terrain steepness. Red = steep, blue = flat. Land in the blue.
- **Coverage:** Maps start blank and fill in as your scanner passes over the surface. A polar orbit at 80° inclination scans everything.

Launch one scanner satellite per body you plan to visit. Even a tiny CubeSat with a scanning antenna and solar panels works. The maps persist across saves.

> SCANsat scanning generates science points passively. A single scanner in polar orbit around the Mun will complete several "Scan [Body]" contracts with zero additional effort.

Wave 0.5 Guide — Building & Planning

Now that you can reach orbit consistently, it's time to go further. Wave 0.5's tools — TWP for planning, DPAI for docking, KAC for mission timing — all come together here.

Going to the Mun

Transfer Burn

From low Kerbin orbit (80–100 km circular, equatorial):

1. Switch to Map view. Rotate the camera so you can see the Mun's orbit.
2. The Mun rises over Kerbin's horizon — when it's about 45 degrees ahead of your craft in its orbit, you're at the transfer window.
3. Create a maneuver node on your orbit. Drag prograde until the projected path intersects the Mun's sphere of influence (SoI). It should take about 860 m/s.
4. Execute the burn at the node. Keep the craft pointed at the maneuver marker on the navball.

> If you miss the Mun's SoI entirely, your burn timing was wrong. Adjust the maneuver node position along your orbit (drag the center circle of the node) — this changes **when** you burn, which changes **where** the Mun is when you arrive.

Mun Capture and Landing

When you enter the Mun's SoI, you'll be on a flyby trajectory. At Mun periapsis (Pe marker), burn retrograde to slow down and enter orbit. Target a 15–20 km circular orbit.

To land:

1. From low Mun orbit, burn retrograde until your trajectory line hits the surface. Target a flat area (the Mun's large craters — maria — are dark, flat regions).
2. As you descend, keep burning retrograde. Watch your surface velocity indicator (click the navball speed readout until it says "Surface").
3. When below 2,000 m altitude, your speed should drop below 100 m/s.
4. In the final 500 m, keep velocity below 20 m/s. Land at less than 6 m/s to avoid breaking anything.
5. SAS set to "Retrograde Hold" (if unlocked) keeps you oriented for the burn — use Stability Assist otherwise and steer manually.

> Kill horizontal velocity first (retrograde marker centered on the horizon line), then control vertical descent. If you tip over on landing, your horizontal velocity was too high.

> **You are standing on another world.**
>
> Four hundred thousand kilometers from home. The sky is black at noon. Kerbin hangs in the sky — a blue-and-white marble that contains everyone you've ever known. It looks small from here. It is small. And you just flew here in a contraption of aluminum and explosives.
>
> Plant the flag. Take a screenshot. Breathe.

Returning from the Mun

1. Launch eastward from the Mun's surface (the Mun rotates slowly — east is still the efficient direction)
2. Establish a low circular orbit (15 km)
3. Burn prograde to escape the Mun — your trajectory will bend back toward Kerbin
4. Target a Kerbin periapsis of 35 km for aerocapture
5. Detach the command pod, deploy parachute, land in the ocean

Basic Docking

Why Dock?

Docking lets you join two spacecraft in orbit. It's essential for Apollo-style Mun missions (leave the lander in Mun orbit, dock to return), space stations, and refueling operations.

Rendezvous

1. Launch the second craft into a lower orbit than the target (faster orbit = catches up) or higher (slower = target catches you).
2. Target the other craft in Map view (click it, select "Set as Target").
3. Create a maneuver node. Adjust until the closest approach markers (orange and magenta arrows) are within 2 km of each other.
4. Execute the burn.
5. When within 2 km, your navball switches to "Target" mode. The prograde marker now shows your velocity **relative to the target**.
6. Burn retrograde (in target mode) to zero out relative velocity.
7. Point at the target (pink circle marker) and burn gently (10–20 m/s) toward it.

8. Repeat: close distance, zero velocity, aim again. Don't rush — docking is a slow dance.

Docking Proper

1. When within 50 m, right-click your docking port and select "Control from Here."
2. Right-click the target's docking port and select "Set as Target."
3. Switch to fine-control mode (Caps Lock — pitch/yaw/roll indicators turn blue).
4. Use RCS (R key) and translation controls (I/J/K/L for up/down/left/right, H/N for forward/back).
5. Align the prograde marker with the target marker on the navball.
6. Drift in at less than 0.5 m/s. The magnetic docking ports will snap together.

> RCS thrusters must be placed symmetrically around your craft's center of mass to avoid unwanted rotation during translation. In the VAB, toggle the center of mass indicator to check. RCS Build Aid (Wave 0.5) makes this much easier.

> **Two ships meet in the dark.**
>
> For a moment, there was empty space between them. Now there is a sealed docking port, and two vessels have become one. You brought them together across kilometers of vacuum, matched velocities to centimeters per second, and touched aluminum to aluminum without a scratch.
>
> If you can dock, you can build anything. Stations. Interplanetary ships. Colonies. Docking is not a skill. Docking is a door.

Space Stations

Now that you can dock, you can build stations.

Why Build a Station?

- **Refueling depot** — dock tankers, mine on Minmus, ferry fuel to orbit
- **Science lab** — the Mobile Processing Lab multiplies science value over time
- **Crew rotation hub** — swap kerbals between missions without landing
- **Relay hub** — high-orbit station with powerful relay antennas extends comms coverage

Station Assembly

1. Launch the core (lab, docking hub, power, probe core)

2. Dock additional modules (fuel tanks, habitation, antennas, extra docking ports)
3. Keep part count reasonable — too many parts and the physics engine crawls. Aim for under 200 parts per station.

> Always include a probe core on every station module. If you undock the wrong port and lose control of a drifting section, a probe core lets you recover it. Also: reaction wheels. Stations in KSP get wobbly without them.

> **You built this.**
>
> It did not exist an hour ago. Now there is a space station in orbit — your space station. Modules you designed, launched, and docked yourself. Solar panels you angled toward the sun. Docking ports waiting for ships that haven't been built yet.
>
> This station wasn't in the game. You put it there. The Kerbol system now contains something that wouldn't exist without you. Every station you build from here adds to a universe that is increasingly, irreversibly yours.

Building Better Rockets

The Rocket Equation

Every rocket design is a tradeoff between three things: payload mass (what you want to deliver), delta-v (how far you want to go), and TWR (how fast you accelerate). Adding more fuel increases delta-v but adds mass, requiring bigger engines to maintain TWR, which adds more mass. This is the tyranny of the rocket equation.

Rules of thumb for efficient designs:

- Each stage should provide roughly 1,500–2,500 m/s of delta-v. More than 3,000 m/s in a single stage wastes mass.
- Liftoff TWR should be 1.3–2.0. Below 1.3 you waste fuel fighting gravity; above 2.0 you risk aerodynamic stress.
- Upper stage TWR can be as low as 0.5 — you're already in space, fighting gravity with orbital velocity.
- For every ton of payload to LKO, expect 4–6 tons of launch vehicle. Better designs push this toward 4.

> KER's VAB readout shows atmospheric and vacuum delta-v. Use **atmospheric** for your first stage (it burns in thick air). Use **vacuum** for upper stages, transfer stages, and landers. Switching mid-design avoids surprises when your 3,400 m/s vacuum-rated first stage can't actually reach orbit.

Staging Strategies

Serial staging (asparagus-adjacent): Boosters feed fuel inward, dropping empty tanks as they go. Most efficient for pure delta-v but complex to build and fly. Used for heavy payloads and interplanetary injection stages.

Parallel staging (boosters + sustainer): Solid or liquid boosters augment a central core. Boosters burn out first and are discarded; the core continues burning. The most common real-world approach — easier to build and more forgiving of asymmetry than serial staging.

SSTO (Single Stage To Orbit): One stage does everything — ascent, circularization, return. Requires RAPIER engines or nuclear thermal rockets in spaceplane configurations. Covered in detail in the Wave 1 spaceplane guide.

Engine Selection

- **Liftoff engines:** High thrust (ASL), moderate Isp. Swivel, Reliant, Skipper, Vector. Mammoth for 5m+ rockets.
- **Upper stage engines:** High vacuum Isp, lower thrust acceptable. Terrier, Poodle, Cheetah, Wolfhound. Vacuum Isp differences matter — a 350s vs 380s Isp upper stage is 8% more delta-v with the same fuel.
- **Deep space engines:** Very high Isp, very low thrust. Nuclear (NERVA), ion (Dawn), or electric (from Near Future Propulsion in Wave 1). Burns measured in minutes to hours.
- **Lander engines:** High TWR for descent braking, good throttle response. Spark, Terrier, Cub. Radial engines for wide landing bases.

Aerodynamics & Stability

Rockets flip during ascent for two reasons: center of drag ahead of center of mass (the dart thrown backward problem), or excessive angle of attack at high dynamic pressure.

Fixes for flipping rockets:
- Add fins at the very bottom. Fins move the center of pressure rearward.
- Make the rocket taller, not wider. A long, thin rocket is more stable than a short, fat one.
- Keep your gravity turn gentle. Below 10 km, stay within 5° of prograde. Above 30 km, the atmosphere is thin enough that aggressive steering is safe.
- Keep speed below 300 m/s while in the thick atmosphere (below 10 km). Faster = more drag = more flipping force.

Designing Aircraft & Spaceplanes

Center of Mass vs Center of Lift

In the SPH, toggle all three indicators (CoM, CoL, CoT). The relationship between these determines whether your plane flies:

- **CoL behind CoM:** Stable. The plane naturally returns to prograde after a disturbance. Good for beginners and long-range cruisers.
- **CoL at CoM:** Neutrally stable. The plane stays where you point it. Requires constant attention but allows maximum maneuverability. Fighter jet territory.
- **CoL ahead of CoM:** Unstable. The plane tries to flip backward. Only flyable with SAS and careful design.

The critical check: drain all fuel tanks in the SPH (right-click each tank, set fuel to zero) and check CoM vs CoL again. Fuel is heavy — your CoM will shift as it burns. If CoM moves **behind** CoL when tanks are empty, your plane will become unstable mid-flight.

> The 1 cause of spaceplane failure: CoM shifts behind CoL as fuel drains. Always check dry-CoM position. If CoM moves backward, move wings slightly aft or add a small fuel tank at the nose to keep the dry CoM forward.

Wing Design

- **Wing area:** More area = more lift at low speed. If your plane needs 150+ m/s just to take off, add wing area.
- **Sweep:** Swept wings reduce supersonic drag. Straight wings are better for subsonic efficiency.
- **Control surfaces:** Elevons (combined elevator + aileron) save parts. Separate pitch, roll, and yaw surfaces give finer control. For spaceplanes, keep control surfaces away from the leading edge (they'll overheat during reentry).

Engine Selection for Aircraft

- **Subsonic cruise (propellers):** Wheesley, Airplane Plus props (Wave 1). Low speed, excellent fuel efficiency. Use for science-gathering planes and early-career contracts.
- **Supersonic jets:** Panther (afterburner for supersonic), Whiplash. Efficient up to Mach 3. Good for high-altitude reconnaissance planes.
- **RAPIER:** The spaceplane workhorse. Air-breathing mode for atmospheric flight; closed-cycle rocket mode for the final push to orbit. RAPIERs auto-switch at 23 km when intake air runs out.
- **Nuclear jets (from NF Aeronautics, Wave 1):** Work in any atmosphere (Duna, Eve, Laythe) — no oxygen needed. Extremely efficient but heavy.

Spaceplane Ascent Profile

1. Accelerate to 140 m/s on the runway, rotate to 10–15°.
2. Climb at 15–20° until 10,000 m. At this altitude, air density drops and drag decreases.
3. At 10,000 m, level off to 5–10°. Accelerate to 1,400 m/s at 20,000 m. This is the RAPIER sweet spot — peak air-breathing thrust.
4. When thrust drops (23–25 km), RAPIERs auto-switch to closed-cycle. Pitch up to 20–30°.
5. Burn to apoapsis > 70 km. Circularize at apoapsis with a small burn (50–200 m/s).
6. Reentry: keep nose at 30–40° pitch. This presents maximum surface area for aerobraking while protecting the cockpit. Don't pitch down — the underside handles heat better than the nose.

Mission Planning

Pre-Launch Checklist

Before launching any mission, verify in the VAB:

1. **Delta-v budget:** KER shows total mission delta-v. Compare against required amounts for your destination.
2. **TWR per stage:** Liftoff stage > 1.3, lander > 1.0 on destination body (cheat: set KER body to Duna to check Duna TWR).
3. **Communications:** Does the craft have an antenna? Can it reach Kerbin from the destination?
4. **Power:** Solar panels or RTG. Check power consumption vs generation in the VAB. If you run out of power, you lose control.
5. **Heat:** Are radiators sufficient if using ISRU or nuclear engines? Converters generate massive heat.
6. **Crew capacity vs crew count:** If you have 3 kerbals and 2 seats, someone's not coming home.

Transfer Window Planning

Transfer windows tell you **when** to burn, not just **how much** delta-v you need.

- **Use TWP:** Open the planner, select origin (Kerbin) and destination (e.g., Duna). The porkchop plot shows delta-v for every departure date. Click the darkest blue region for the most efficient window.
- **Phase angles (eyeball method):** Duna should be 44° ahead of Kerbin in its orbit. Eve 54° behind. Jool 96° ahead. If you can't use a planner, these rough angles get you close.
- **Depart from low circular orbit:** A 100 km circular orbit is ideal for interplanetary ejection. Elliptical orbits complicate ejection-angle calculations.

- **Mid-course correction:** About halfway to your destination, create a small correction burn (10–50 m/s) to fine-tune your encounter periapsis. This is far cheaper than correcting during the departure burn.

> Duna is the best first interplanetary target. Its atmosphere (thin, but enough for parachutes), low gravity (0.3g), and moderate transfer cost (1,100 m/s from LKO) make it forgiving. Ike is even easier — no atmosphere but ultra-low gravity and it's right there when you arrive at Duna.

Where to Go from Here

You can now land on the Mun, dock spacecraft in orbit, and assemble space stations. You've mastered the core skills of orbital operations. When you're comfortable with rendezvous and can plan Mun missions without step-by-step instructions, you're ready for Wave 1 — interplanetary travel, ISRU mining, and exploring the expanded solar system.

Player Challenges — Wave 0.5

Test your Mun, docking, and station skills.

- **Single-Stage Lander:** Mun landing and return with no staging during descent or ascent. One engine, one tank, one landing.
- **Apollo Style:** Design a two-part mission — Command/Service Module stays in Mun orbit while a separate lander descends. Dock them back together before returning.
- **No RCS Docking:** Dock two craft using only main engine thrust. No RCS thrusters allowed. Translation via careful tapping.
- **Three-Module Station:** Launch and dock three separate modules (core + hab + fuel depot) into a single station. No module may exceed 30 parts.
- **Polar Mun Landing:** Land on the rim of a Mun polar crater. The terrain is steep, the lighting eternal twilight. Good luck.
- **Slope Landing:** Land on Minmus in the Slopes biome. Inclination 15–30°. Your lander must not tip.

[WAVE 1]

[Going Further]

[Graphics, Parts & Expanded Horizons]

Wave 1 — Mod List

About Wave 1

Wave 1 expands KSP with graphics overhauls, parts packs, and light mechanical additions that add content without fundamentally changing difficulty. You should be comfortable with the skills taught in Waves 0–0.5 (orbit, Mun landings, docking, space stations) before installing these mods.

Criterion	Wave 0	Wave 0.5	Wave 1	Wave 2
Gameplay changed?	No	No	Adds content	Adds complexity
New parts?	No	No	Yes (2000+)	Yes (colony + interstellar)
Learning curve	None	Tools & planning	Parts, planets, graphics	Life support, colonies, FTL
Resource chains?	No	No	Fuel types (CRP/cryo)	Full (LS, colony, reactor fuel)
Failure conditions?	No (revert only)	No (revert only)	Mission failure (no revert)	Permanent (dead kerbals)
Progression goal	Stable orbit	Mun, docking, stations	Interplanetary	Interstellar, self-sustaining
Save-breaking?	No	No	Unlikely	Likely (choose wave carefully)

> Wave 1 mods are cumulative with Waves 0–0.5. Do not skip the earlier waves — Wave 1 includes their mods. Install Waves 0 and 0.5 first, play until you've mastered orbital operations and station assembly, then add Wave 1.

Mod List

MechJeb

Impact: UI, Mechanics

Version: Compatible with KSP 1.12.x.

Advanced autopilot and flight computer. Provides automated ascent, rendezvous, landing, maneuver planner, and a customizable data readout. Useful as a teaching tool — watch what MechJeb does, then learn to do it yourself.

Adds an autopilot module part and data windows. The autopilot can fly entire missions autonomously. Purely additive — you can ignore the autopilot and use it only for the readouts, which are similar to KER.

AtmosphereAutopilot

Dependencies: ModuleManager

Impact: UI, Mechanics

Version: Compatible with KSP 1.12.x. Does not conflict with MechJeb — AA handles atmospheric flight, MJ handles spaceflight. Use both for complete autopilot coverage.

Fly-by-wire flight control system with automatic trim and advanced autopilot modes. Replaces KSP's basic SAS with proper control-surface coordination, autotrim, and stability augmentation. Includes autopilot modes for altitude hold, heading hold, and autothrottle — makes long-duration atmospheric flight hands-off.

Improves atmospheric flight control with coordinated surface deflection and automatic trim. The autopilot modes handle altitude, heading, and speed — useful for circumnavigation flights and long-duration atmospheric surveys. Complements MechJeb (space autopilot) with atmospheric specialization.

Scatterer

Dependencies: ModuleManager

Impact: Graphics

Version: Compatible with KSP 1.12.x. Installed automatically with AVP. Blackrack's Volumetric Clouds requires Scatterer for its lighting integration. Configurable via the Scatterer toolbar menu in flight.

Atmospheric scattering, ocean shaders, and sunflare effects. Gives planets proper atmospheric haze, Rayleigh scattering (blue skies on Kerbin), and reflective water surfaces with wave animation. By Blackrack — also the author of Volumetric Clouds. Installed automatically as an AVP dependency.

Replaces KSP's flat atmospheric rendering with physically-based scattering. Oceans gain reflections and wave animation. Sunflare is customizable with multiple presets. Moderate GPU impact — adjust ocean and atmosphere quality in settings.

Astronomer's Visual Pack

Dependencies: EnvironmentalVisualEnhancements, Scatterer, ModuleManager

Impact: Graphics

Comprehensive visual overhaul: high-resolution skybox, revamped cloud layers (via EVE), and atmospheric scattering configs (via Scatterer). The gold standard for KSP visuals.

Changes skybox, cloud textures, and atmospheric scattering. No gameplay impact — purely visual. Performance impact depends on texture resolution chosen during install.

Version: Compatible with KSP 1.12.x. Requires 8 GB+ VRAM for high-res textures.

Deferred

Dependencies: ModuleManager

Impact: Graphics

Version: Compatible with KSP 1.12.x.

Replaces KSP's forward renderer with a deferred renderer. Enables better lighting (many dynamic lights without performance collapse), physically-based shaders, and improved shadow handling.

Changes the rendering pipeline. More dynamic lights, better reflections, and many of the features Planetshine provides natively. Significant visual improvement with a smaller performance hit than stacking multiple lighting mods.

Parallax Continued

Dependencies: Parallax-StockTextures, ModuleManager

Impact: Graphics

Version: Compatible with KSP 1.12.x. Requires a DX11-capable GPU.

Replaces stock terrain scatter with high-detail tessellated ground textures: rocks, grass, trees, and surface detail at close range. Transforms planetary surfaces from flat painted terrain into detailed 3D landscapes.

Adds GPU-intensive terrain tessellation. Near-ground visuals are dramatically improved. Requires a GPU capable of handling tessellation shaders. Use Parallax-StockTextures for stock planets; additional texture packs needed for OPM or Kcalbeloh.

Vapor Cones

Impact: Graphics

Version: Compatible with KSP 1.12.x.

Adds supersonic vapor cone effects around your craft when breaking the sound barrier in atmosphere. The condensation cloud that forms during transonic flight.

Purely visual — no gameplay effect. Triggers when your vessel exceeds Mach 1 in sufficient atmospheric density.

Reentry Particle Effect Renewed (Firefly)

Adds plasma trail and spark effects during atmospheric reentry. Heat shields and leading edges glow with realistic ablation particles.

Dependencies: ModuleManager

Impact: Graphics

Version: Compatible with KSP 1.12.x.

Visual only — reentry heating damage is unchanged. Enhances the drama of atmospheric entry with particle trails and incandescent effects.

Texture Replacer Replaced

Impact: Graphics

Version: Compatible with KSP 1.12.x.

Replaces stock textures: kerbal suits, heads, skybox, and EVA visor reflections. The modern maintained fork of the original TextureReplacer.

Enables custom kerbal suits, skybox replacement, and visor reflections. Purely visual. Requires downloading or creating texture packs — the mod itself is a framework.

TUFX (Textures Unlimited FX)

Dependencies: ModuleManager

Impact: Graphics

Version: Compatible with KSP 1.12.x. Fox's Experimental profile is a manual download, not on CKAN.

Post-processing framework: ambient occlusion, bloom, anti-aliasing, color grading, HDR tonemapping. Includes several built-in profiles. For a cinematic, desaturated look, try Fox's Experimental profile (manual install from KSP forums).

Adds a post-processing stack to the flight and editor scenes. Profiles are toggleable in-game via toolbar. Performance impact depends on profile — ambient occlusion and high-quality AA have the largest cost.

Environmental Visual Enhancements (EVE)

Dependencies: ModuleManager

Impact: Graphics

Version: Compatible with KSP 1.12.x. Installed automatically with AVP. Blackrack's Volumetric Clouds is a paid Patreon mod that replaces EVE's cloud shader — manual install from Blackrack's Patreon.

Cloud and atmospheric effects framework. Adds volumetric cloud layers, city lights on the dark side of Kerbin, and atmospheric glow. The base framework that Astronomer's Visual Pack, Spectra, and other visual packs build their cloud configs on top of. Installed automatically as a dependency of AVP.

Provides the cloud rendering engine. Does nothing on its own without config packs (AVP, Spectra, BoulderCo). AVP includes EVE as a dependency and provides configs. Blackrack's Volumetric Clouds (Patreon) upgrades EVE's cloud rendering to a full volumetric system.

Blackrack's Volumetric Clouds

Dependencies: EnvironmentalVisualEnhancements, Scatterer, ModuleManager

Impact: Graphics

Version: Compatible with KSP 1.12.x. Manual install from Blackrack's Patreon (free — Google Drive download link on the post page). Download the latest release, extract to GameData/. Worth supporting Blackrack on Patreon if you can.

True volumetric cloud rendering for KSP. Replaces EVE's 2D cloud layers with fully 3D volumetric clouds that react to lighting, cast shadows, and have real depth. Transforms atmospheric flight — clouds are no longer flat textures but massive 3D formations you can fly through. A Patreon-supported mod by Blackrack (also the author of Scatterer).

Upgrades EVE's cloud system to volumetric rendering. Significant visual improvement — clouds have real 3D volume, self-shadowing, and dynamic lighting. Performance impact is moderate to high depending on cloud quality settings. Requires EVE and Scatterer (both already installed via AVP).

Distant Object Enhancement

Dependencies: ModuleManager

Impact: Graphics

Version: Compatible with KSP 1.12.x. Works with all planet packs (OPM, Kcalbeloh). Vessel flares use the same sky-dimming system — disable if you find distant ship flares distracting.

Renders planets and moons as visible points of light in the sky, with proper brightness and color based on distance and phase. Distant vessels appear as flares against the skybox. Vastly improves the sense of scale — you can see Jool from Kerbin orbit, Duna as a red dot, and your space station winking in the sunlight.

Adds dynamic sky-dimming and distant vessel rendering. Planets are visible even when they're just a few pixels across. No performance cost — uses efficient point-sprite rendering.

Textures Unlimited

Dependencies: ModuleManager

Impact: Graphics

Version: Compatible with KSP 1.12.x. Framework mod — no visual change on its own. Install TURD (Textures Unlimited Recolour Depot) or TU-compatible texture packs to see PBR effects.

PBR (Physically Based Rendering) shader framework for KSP parts. Enables metallic, reflective, and roughness-based textures on supported parts. A separate concept from TextureReplacer — TU changes how parts render, while TR replaces texture files. Required by some parts mods for their reflective/textured materials.

Adds PBR shader support to KSP's rendering pipeline. Parts default to standard KSP shaders — only mods that ship TU-compatible textures use PBR. Required by Textures Unlimited Recolour Depot (TURD) for in-editor part recoloring.

Textures Unlimited Recolour Depot (TURD)

Dependencies: TexturesUnlimited, ModuleManager

Impact: Graphics, UI

Version: Compatible with KSP 1.12.x. Manual install from forum thread — not on CKAN. Requires TexturesUnlimited. TURD for DLC, B9, and MkIV recolor packs are available on the same forum thread for additional part support.

In-editor part recoloring using Textures Unlimited's PBR shaders. Select any supported part in the VAB/SPH and recolor it with preset palettes or custom colors — make your rockets any color you want while maintaining the PBR material quality. Works with stock parts and mod packs that ship TURD configs.

Adds a part recoloring GUI in the editor. Parts must have TURD-compatible textures — stock parts and many mod parts have community configs available. The recolor is saved with the craft file and visible in flight. Pure visual — no gameplay impact.

Simple Repaint

Dependencies: TexturesUnlimited, ModuleManager

Impact: Graphics

Version: Compatible with KSP 1.12.x. Requires TexturesUnlimited. Complements TURD — install both for maximum part customization coverage.

Part re-shader for parts that lack TURD support. Provides basic color/reflectivity adjustments for mod parts that don't have full PBR texture configs — a lighter-weight alternative to TURD for quick visual customization. Complements TURD by covering parts TURD doesn't support.

Adds basic shader-based repainting for parts without TURD configs. Uses TexturesUnlimited's PBR framework. Not a replacement for TURD — use both: TURD for parts with full configs, SimpleRepaint for everything else.

Shabby / Shaddy

Dependencies: TUFX

Impact: Graphics

Version: Compatible with KSP 1.12.x. Both Shabby and Shaddy (CKAN: "Shabby", "Shaddy") install together as a profile pack. Select them from TUFX's in-game profile selector.

TUFX post-processing profiles by benjee10. Shabby provides a clean, cinematic profile suitable for screenshots and general gameplay. Shaddy extends Shabby with more profile variants. Install via CKAN and select the profile from the TUFX toolbar menu in flight.

Adds additional TUFX profiles beyond the built-in defaults. Pure configuration — toggles between profiles in-game with no performance cost beyond the post-processing effects themselves.

Pood's Skyboxes

Impact: Graphics

Version: Compatible with KSP 1.12.x. Manual install from forum thread — not on CKAN. Download, choose your preferred variant, place in GameData/. TextureReplacerReplaced can also install skybox textures.

High-resolution skybox replacements by Poodmund. Includes three variants: Milky Way (detailed galaxy panorama), Calm Nebula (subtle, warm-toned nebula — the personal favourite), and Deep Star Map (dense starfield for deep-space immersion). Replaces KSP's default flat galaxy background with rich, detailed space vistas.

Replaces the skybox cubemap texture. No performance impact. Purely visual. Three variants available in one download — Calm Nebula is recommended for its warm, atmospheric feel that complements AVP and Scatterer.

Restock

Dependencies: ModuleManager

Impact: Graphics, Parts

Version: Compatible with KSP 1.12.x.

Complete visual revamp of every stock part: models, textures, and effects. Maintains stock dimensions and attachment points — craft files are 100% compatible. Makes stock parts look like they belong in a modern game.

Replaces art assets for all stock parts. No functional changes — same mass, cost, tech tree placement, and attachment nodes. Existing craft files work unchanged. Pair with Restock+ for additional parts.

⚠ Conflicts: Any mod that relies on the visual appearance of specific stock parts (e.g., some part-welding mods) may need patches.

Restock+

Dependencies: ReStock, ModuleManager

Impact: Parts

Version: Compatible with KSP 1.12.x. Requires Restock.

Adds new stock-alike parts that fill gaps in the stock lineup: larger reaction wheels, additional fuel tank sizes, resized adapters, and more. All parts follow Restock's visual quality.

Adds new parts in stock sizes and tech nodes. Fills gaps rather than adding new mechanics — a 2.5m reaction wheel, 0.625m RCS tanks, missing adapter pieces, etc.

Waterfall

Dependencies: ModuleManager

Impact: Graphics

Modern, mesh-driven engine plume framework. Replaces stock particle-based exhaust with continuous, physically-inspired plumes that expand realistically in vacuum and contract under atmospheric pressure. Includes configs for stock engines.

Version: Compatible with KSP 1.12.x. WaterfallRestock provides configs for ReStock engines.

Replaces engine exhaust visuals. No performance or gameplay changes. Stock engine plumes are significantly improved out of the box; additional configs available for mod engines.

Waterfall for Restock

Dependencies: Waterfall, ReStock

Impact: Graphics

Version: Compatible with KSP 1.12.x. Requires both Waterfall and Restock.

Patches Waterfall engine effects onto Restock's revamped engine models. Ensures the beautiful Restock engine models also get beautiful Waterfall plumes.

Bridges Waterfall and Restock so that Restock's engine models use Waterfall's plume system instead of falling back to stock particles.

Restock Waterfall Expansion (RSMP)

Dependencies: WaterfallRestock, Waterfall, ReStock, ModuleManager

Impact: Graphics

Version: Compatible with KSP 1.12.x. Requires Waterfall, WaterfallRestock, and ReStock. CKAN automatically resolves dependencies — just install and it works.

Additional Waterfall plume configurations for ReStock engines beyond what WaterfallRestock provides. Adds plumes to more niche ReStock engines, RCS thrusters, and SRBs. Install alongside WaterfallRestock for complete plume coverage on all ReStock parts. The mod formerly known as RSMP (ReStock Plume Expansion).

Extends Waterfall plume coverage to engines and parts that WaterfallRestock doesn't cover. Pure graphics — no gameplay changes. May cause minor issues with specific engine configs; CKAN handles the dependency chain automatically.

Rocket Sound Enhancement

Dependencies: ModuleManager

Impact: Graphics

Version: Compatible with KSP 1.12.x. Complements Waterfall's visual plumes with matching audio quality. Default configs cover stock engines; additional

Replaces stock engine sounds with realistic, high-quality audio. Engine roar changes with throttle, distance, and atmospheric density. Includes sonic boom effects — when the camera pans ahead of a supersonic craft, sound cuts out (you're traveling faster than sound). Transforms the auditory experience of launches and atmospheric flight.

Replaces stock sound effects with layered, dynamic audio. Engine sounds respond to throttle position and camera distance. Includes realistic sonic-boom physics and at-configs available for mod engines.

mospheric sound attenuation. Pure audio — no gameplay changes.

Community Tech Tree

Dependencies: ModuleManager

Impact: Mechanics, Gameplay

Version: Compatible with KSP 1.12.x. Most parts mods (Near Future, Planetside, etc.) have built-in CTT support. Recommended but not required for Hide Empty Tech Tree Nodes (Wave 0).

Expands the stock tech tree with additional nodes for modded parts. Instead of mods cramming their advanced parts into the stock tree's last few nodes, CTT provides dedicated progression nodes for nuclear propulsion, colonization, life support, and deep-space technologies. The foundation for organized modded career progression.

Restructures and extends the tech tree with dozens of new nodes. Changes career/science mode progression by spreading parts across a deeper, more specialized tree. Mod parts that support CTT will auto-assign to appropriate new nodes. Does not affect Sandbox mode.

⚠ Conflicts: Other tech tree mods (Engineer Tech Tree, Un-Kerballed Start, etc.) are incompatible — choose one tech tree for your save. Stock parts are unaffected and remain in their original nodes.

Fuel System

Many parts mods in Wave 1 use custom fuel types beyond stock LiquidFuel/Oxidizer. The mods below define and manage these resources so engines and tanks from historical, modern, and shuttle packs use their real-world propellants. Install these once — all dependent parts mods will auto-configure.

Community Resource Pack

Impact: Mechanics

Version: Compatible with KSP 1.12.x. Required by Bluedog DB, Tantares, Tundra, Artemis CK, HabTech2, and many others. Install once.

Defines shared resource definitions used by the KSP modding ecosystem. Adds Hydrogen, Methane, Kerosene, Hydrazine, and dozens of other resources to the game. No parts or gameplay on its own — other mods reference these definitions for their engines and tanks.

Registers new resource types in the game database. Required by most parts packs that use custom fuels. Does nothing visible by itself — it's a shared library that other mods depend on.

B9 Part Switch

Dependencies: ModuleManager

Impact: Mechanics

Version: Compatible with KSP 1.12.x. Required by virtually every parts mod that uses custom fuel types.

Allows fuel tanks to switch between different resource configurations in the editor. A single tank can hold LiquidFuel/Oxidizer (stock), LH2/Oxidizer (cryogenic), Methane/Oxidizer, or monopropellant — switchable with a dropdown in the VAB. Foundation for the entire custom-fuel ecosystem.

Adds part-switching capability to fuel tanks. The tank model doesn't change — only the resources it contains. Required by most parts mods that use non-stock fuels. Can also switch textures and models on supported parts.

⚠ Conflicts: InterstellarFuelSwitch and Firespitter's fuel switch provide overlapping functionality. B9PartSwitch is the modern standard — only install alternatives if a specific mod requires them.

Cryogenic Tanks

Dependencies: CommunityResourcePack, B9PartSwitch, ModuleManager

Impact: Parts, Mechanics

Version: Compatible with KSP 1.12.x. Boil-off is configurable — you can disable it entirely if you prefer simpler fuel management.

Adds cryogenic fuel tank options and boil-off mechanics. LH2 and Methane require insulated tanks — they slowly evaporate if stored in standard tanks. Includes active cooling parts that eliminate boil-off at the cost of ElectricCharge.

Adds insulated tank variants with boil-off management. LH2 tanks lose fuel over time unless actively cooled. This is a light resource management mechanic — not a failure condition, but something you need to account for on long missions.

Cryogenic Engines

Dependencies: CommunityResourcePack, CryoTanks, B9PartSwitch, ModuleManager

Impact: Parts, Mechanics

Version: Compatible with KSP 1.12.x. Best paired with Near Future Electrical for powering active cooling on long missions.

Adds LH2/Oxidizer-fueled rocket engines in the Near Future style. Higher Isp than stock chemical engines but requires managing cryogenic fuel boil-off. Includes upper-stage and deep-space engines optimized for efficiency over thrust.

Adds LH2-burning engines that integrate with the CryoTanks boil-off system. Engines range from 1.25m upper-stage to 3.75m heavy-lift cryogenic. Part of the Near Future Technologies family of mods.

Deployable Engines

Dependencies: ModuleManager

Impact: Mechanics

Version: Compatible with KSP 1.12.x. Required by Rocket Motor Menagerie, Artemis Construction Kit, and CryoEngines.

Engine animation framework. Allows engine nozzles to extend, retract, and deploy with animations. Required by Rocket Motor Menagerie, CryoEngines, Artemis Construction Kit, and other mods that feature animated engine parts. No standalone effect.

Enables deploy/retract animations on supported engines. Pure framework — does nothing unless another mod provides animated engine parts. Required dependency for several Wave 1 parts packs.

Nertea Suite Extensions

Heat Control

Dependencies: ModuleManager

Impact: Parts, Mechanics

Version: Compatible with KSP 1.12.x. System Heat recommended. Both work together: System Heat = framework, Heat Control = parts.

Advanced radiator parts by Nertea: deployable radiators, graphene panels, and active cooling systems for managing extreme heat from nuclear reactors and engines. System Heat provides the thermal framework — Heat Control provides the radiator parts.

Adds high-temperature radiator parts. Essential for nuclear reactors (NF Electrical) and nuclear engines (Kerbal Atomics). System Heat recommended for advanced thermal simulation that uses these radiators.

System Heat

Dependencies: ModuleManager

Impact: Mechanics

Version: Compatible with KSP 1.12.x. Recommends Heat Control. Skip System Heat if you prefer simpler thermal management (stock core-heat with Heat Control radiators).

Advanced thermal simulation by Nertea. Replaces KSP's simplified core-heat with detailed heat flow, thermal mass, and warm-up/cool-down phases for nuclear reactors. Adds realistic thermal management — not just "add radiators" but manage heat distribution and temperature limits.

Overhauls core-heat with detailed simulation. Nuclear reactors and ISRU converters have real thermal requirements — manage heat flow, temperature, and thermal inertia. Recommends Heat Control for compatible radiator parts designed for this system.

Kerbal Atomics

Nuclear thermal rockets by Nertea. NERVA-style engines with LH2 propellant and 800–1,000s Isp.

Dependencies: CommunityResourcePack, CryoTanks, B9PartSwitch, DeployableEngines, ModuleManager

Impact: Parts, Mechanics

Version: Compatible with KSP 1.12.x. Requires CryoTanks and CRP. Pairs with NF Electrical for reactor power and Heat Control/System Heat for thermal management.

Includes open-cycle gas-core and closed-cycle nuclear engines — the bridge between chemical rockets and fusion drives. Uses CryoTanks for LH2 boil-off management.

Adds nuclear thermal engines consuming LH2 for high-efficiency deep-space propulsion. Requires cryogenic fuel management. Higher Isp than chemical engines at the cost of engine mass and radiation. Fits between NF Propulsion and FFT in the Nertea engine progression.

Space Dust

Dependencies: ModuleManager

Impact: Mechanics, Gameplay

Version: Compatible with KSP 1.12.x. Required for full FFT gameplay. Without SpaceDust, exotic FFT fuels are KSC-only. With it, fuel acquisition becomes a mission objective itself.

Exotic resource harvesting framework by Nertea. Places harvestable antimatter, fusion fuel, and exotic particles in specific orbital bands, planetary rings, and atmospheric layers. Required for FFT's advanced fuel chains — send dedicated harvester missions to collect resources rather than buying them at KSC.

Defines harvestable exotic resource locations — specific orbits, rings, and atmospheres. Transforms FFT's fuel chain from KSC purchases to in-situ harvesting expeditions. Purpose-built harvester parts needed per resource type.

Heavy Lift

Hullcam VDS Continued

Dependencies: ModuleManager

Impact: Parts, UI

Version: Compatible with KSP 1.12.x.

Adds functional camera parts: launch pad cameras, docking cameras, rover cameras, and telescope lenses. View the game world through any camera part for cinematic shots or improved situational awareness during docking and landing.

Adds camera parts at various tech nodes. Each camera provides a live feed view in a window or fullscreen. No gameplay changes — purely a new way to see what's happening around your craft.

Near Future Solar

Expanded solar panel selection: curved arrays, blanket panels, concentrator photovoltaics, and giant or-

Dependencies: NearFutureProps, ModuleManager

Impact: Parts

Version: Compatible with KSP 1.12.x.

bital arrays. Solar power that scales from tiny probes to massive space stations.

Adds solar panel parts at multiple tech tiers. All produce ElectricCharge — no new resources. Panels range from small probe-sized to station-scale deployable arrays.

Near Future Electrical

Dependencies: NearFutureProps, CommunityResourcePack, DynamicBatteryStorage, ModuleManager

Impact: Parts, Mechanics

Version: Compatible with KSP 1.12.x. Reactor fuel is a one-time load — no refueling chain required.

Nuclear reactors, fission generators, capacitors, and high-capacity batteries for deep-space missions where solar panels produce negligible power. Includes a reactor management UI for monitoring core temperature and fuel consumption.

Adds reactors that consume EnrichedUranium and produce ElectricCharge, plus capacitors for burst power. Introduces a new resource (EnrichedUranium) and waste heat management via radiators. Scales to end-game power demands.

Near Future Propulsion

Dependencies: NearFutureProps, NearFutureElectrical, CommunityResourcePack, ModuleManager

Impact: Parts, Mechanics

Version: Compatible with KSP 1.12.x. Engines require Near Future Electrical or equivalent power sources.

Advanced electric and nuclear propulsion: VASIMR-style plasma thrusters, magnetoplasmadynamic engines, pulsed inductive thrusters, and colloid ion engines. High Isp, low thrust — for deep-space efficiency, not launch power.

Adds engines that consume vast amounts of ElectricCharge (and sometimes Argon, Xenon, or Lithium) in exchange for extremely high Isp. Requires powerful electrical infrastructure (NF Electrical or massive solar) to operate.

Near Future Construction

Dependencies: NearFutureProps, ModuleManager

Impact: Parts

Large structural parts: octo-girders, hexagonal trusses, docking connectors, and modular construction components for building orbital stations and interplanetary vessels in space.

Adds truss and structural parts. No new mechanics — just more ways to build large space structures without excessive part counts.

Version: Compatible with KSP 1.12.x.

Near Future Spacecraft

Dependencies: NearFuture-Props, ModuleManager

Impact: Parts

Version: Compatible with KSP 1.12.x.

Advanced command pods and service modules: 3-kerbal capsules, 7-kerbal orbital modules, deep-space habitation pods, and integrated RCS/service bays. Pods designed for long-duration crewed missions.

Adds crewed parts with larger capacities and integrated features (built-in RCS, experiments, storage). Pure parts addition — no new mechanics.

Near Future Launch Vehicles

Dependencies: NearFuture-Props, ModuleManager

Impact: Parts

Version: Compatible with KSP 1.12.x.

Heavy-lift engines, 5m and 7.5m fuel tanks, large SRBs, and engine mounting plates for building Saturn V-class and larger rockets. Extends the stock rocket scale upward.

Adds large-diameter parts (5m–7.5m) and high-thrust engines. No new mechanics — these are bigger versions of stock rocket components for launching heavier payloads.

Near Future Aeronautics

Dependencies: NearFuture-Props, ModuleManager

Impact: Parts, Mechanics

Version: Compatible with KSP 1.12.x. Nuclear jets require a nuclear reactor from this pack or Near Future Electrical.

Advanced aircraft parts: nuclear thermal turbojets, multi-mode atmospheric engines, large wings, and intake systems. Extends atmospheric flight with engines that work on planets without oxygen (Duna, Eve, Laythe's atmosphere).

Adds nuclear jet engines that function in any atmosphere (not just oxygenated). Uses IntakeAtm — these engines heat atmospheric gas rather than burning it, enabling powered flight on Duna or Eve.

Mk2 Expansion

Dependencies: ModuleManager

Triples the Mk2 fuselage part catalog: inline docking ports, cargo ramps, crew cabins, science labs, RCS-integrated sections, nuclear reactors, and more. Every Mk2-shaped thing you've ever wanted.

Impact: Parts

Version: Compatible with KSP 1.12.x.

Adds Mk2-profile parts at various tech nodes. All parts maintain the stock Mk2 cross-section and attachment rules. No new mechanics.

SCANsat

Dependencies: ModuleManager

Impact: Parts, Mechanics, UI

Version: Compatible with KSP 1.12.x. Does not alter stock resource scanning — SCANsat scanning is parallel and additive.

Adds surface-scanning parts and a map viewer. Launch scanning satellites to map planetary biomes, altimetry, ore concentrations, and anomaly locations. Maps are persistent and reveal detail as scanning coverage increases.

Adds scanner parts (RADAR, SAR, multispectral, biome) and a Map view accessible from the toolbar. Scanning requires placing satellites in appropriate polar orbits. Maps provide biome and terrain data useful for landing site selection and resource prospecting.

FreeIva

Dependencies: ModuleManager

Impact: Mechanics, UI

Version: Compatible with KSP 1.12.x. Works with most stock and mod IVAs. Some mod IVAs may not have passable colliders defined.

Lets kerbals walk freely inside crewed parts in IVA view. Move between connected habitable modules, float through passageways, and explore the interiors of your spacecraft in first person.

Adds first-person movement inside IVAs. Habitable parts must be connected via passable nodes (docking ports, crew tubes). Adds immersion without affecting flight mechanics. Kerbals cannot EVA from IVA — exiting still goes through the normal EVA button.

Historical Rockets

These mods add meticulously detailed replicas of real-world spacecraft and launch vehicles from the early space age through the Apollo era. Parts are balanced for stock-scale KSP (2.5× smaller than real life) and integrate with the Community Tech Tree and CRP fuel system.

Bluedog Design Bureau

Dependencies: B9PartSwitch, CommunityResourcePack,

Comprehensive US rocketry pack covering Mercury, Gemini, Apollo, and dozens of historical launchers (Atlas, Titan, Delta, Scout). Hundreds of parts: command pods, service modules, lunar landers, engines, fuel tanks, and science instruments — all in a unified

ModuleManager, DeployableEngines

Impact: Parts, Mechanics

Version: Compatible with KSP 1.12.x. Requires CRP, B9PartSwitch, and DeployableEngines. Tantares is the Soviet counterpart — both are designed to work together.

stockalike art style. The definitive historical US space program mod.

Adds a complete US historical spacecraft and launcher catalog. Engines use real-world propellants (Kerosene/LOX, LH2/LOX, hypergolics) via CRP. Tech tree placements roughly follow historical chronology. Pairs with Tantares for the full Cold War space race experience.

Tantares — Soviet Spacecraft

Dependencies: B9PartSwitch, ModuleManager

Impact: Parts, Mechanics

Version: Compatible with KSP 1.12.x. Pair with NewTantaresLV for Soviet launchers (Soyuz rocket, Proton, N1). TantaresSP adds Soviet interplanetary probes.

Soviet crewed spacecraft: Vostok, Voskhod, Soyuz, Progress, TKS/VA, and Salyut/Mir station modules. The definitive Soviet space program parts pack in stockalike style. Includes command pods, orbital modules, service modules, docking systems, and station parts.

Adds Soviet crewed spacecraft parts. Recommends NewTantaresLV for launch vehicles and TantaresSP for uncrewed probes. Engines use real-world propellants via CRP. Designed to complement BluedogDB for a complete Cold War era mod set.

⚠ Conflicts: The original Tantares (deprecated) conflicts with NewTantares — install NewTantares only.

Tantares LV — Soviet Launchers

Dependencies: B9PartSwitch, ModuleManager

Impact: Parts, Mechanics

Version: Compatible with KSP 1.12.x. Recommended companion to NewTantares.

Soviet launch vehicles: Soyuz rocket family, Proton, N1, and Energia. Fuel tanks, engines, and fairings designed to match the Tantares spacecraft parts. Provides the rockets that carry Vostok, Soyuz, and TKS spacecraft to orbit.

Adds Soviet launcher parts. Intended for use with NewTantares — the rockets are scaled and balanced to carry Tantares payloads. Engines use real-world propellants via CRP.

Tantares SP — Soviet Space Probes

Dependencies: B9PartSwitch, ModuleManager

Soviet uncrewed probes and interplanetary spacecraft: Luna, Venera, Mars, and other historical Soviet probe programs. Probe cores, science instruments, and antenna parts in the Tantares art style.

Impact: Parts

Version: Compatible with KSP 1.12.x. NeptuneCamera (recommended) adds camera parts that integrate with probe cores.

Adds Soviet probe parts for uncrewed exploration. Complements NewTantares and NewTantaresLV. NeptuneCamera recommended for integrating camera functionality with probe parts.

Modern Rockets

Eisenhower Astronautics

Dependencies: B9PartSwitch, CommunityResourcePack, ModuleManager

Impact: Parts, Mechanics

Version: Compatible with KSP 1.12.x. Complements Tantares by providing modern Russian vehicles alongside the Soviet-era Tantares parts.

Modern Russian spacecraft and launch vehicles: Angara rocket family, modern Soyuz variants, and contemporary Russian space hardware. Extends the Russian/Soviet lineage from Tantares into the 21st century with updated designs and modern engine configurations.

Adds modern Russian rocket and spacecraft parts. Uses CRP for propellant configuration. Shares Tantares' design language but with modern vehicle profiles.

Tundra Exploration

Dependencies: B9PartSwitch, CommunityResourcePack, ModuleManager, DeployableEngines

Impact: Parts, Mechanics

Version: Compatible with KSP 1.12.x. Tundra Technologies adds probe buses and satellite parts that complement the Falcon/Dragon lineup.

SpaceX-inspired parts: Falcon 9, Falcon Heavy, Dragon capsule, and Starship. Includes grid fins, landing legs, and superdraco thrusters. Build and fly reusable rockets with propulsive landing capability — Falcon-style booster recovery and Dragon-style capsule landings.

Adds SpaceX-inspired launchers and spacecraft. Engines use Kerosene/LOX (Falcon) and Methane/LOX (Starship) via CRP. Landing legs, grid fins, and RCS pods enable propulsive recovery operations.

Tundra Technologies

SpaceX-inspired probe buses, satellite platforms, and uncrewed spacecraft parts. Adds modular satellite buses, probe cores, and instrument platforms

Dependencies: B9PartSwitch, ModuleManager

Impact: Parts

Version: Compatible with KSP 1.12.x. Designed to pair with Tundra Exploration for the full SpaceX-style ecosystem.

designed to pair with Tundra Exploration's Falcon launchers.

Adds satellite and probe parts in the Tundra art style. Complements Tundra Exploration for building complete Falcon-launched satellite missions.

Shuttle, SLS & ISS

Artemis Construction Kit

Dependencies: B9PartSwitch, CommunityResourcePack, ModuleManager, Benjee10-SharedAssets, DeployableEngines, AnimatedDecouplers, SimpleAdjustableFairings, HabTechProps

Impact: Parts, Mechanics

Version: Compatible with KSP 1.12.x. Requires Breaking Ground DLC for robotic parts support. Rocket Motor Menagerie adds RS-25 engine variants.

SLS rocket and Orion spacecraft parts. Build NASA's modern deep-space launch system: core stage with RS-25 engines, SRBs, Orion capsule, European Service Module, and ICPS upper stage. Recommended SLS/Artemis mod for this modlist.

Adds SLS and Orion parts. RS-25 engines use LH2/LOX via CRP. SRBs use solid fuel. Orion has integrated RCS, parachutes, and heat shield. Designed to reach the Moon and beyond. Rocket Motor Menagerie adds RS-25 variants for more engine options.

Rocket Motor Menagerie

Dependencies: B9PartSwitch, CommunityResourcePack, ModuleManager, DeployableEngines

Impact: Parts, Graphics

Version: Compatible with KSP 1.12.x. Waterfall and CryoTanks recommended for full visual and mechanical integration. DeployableEngines required for nozzle animations.

Expanded RS-25 (SSME) engine variants with deployable nozzles and Waterfall plumes. Adds multiple RS-25 configurations — standard, high-expansion-ratio, and extended-nozzle variants for different flight regimes. Complements Artemis Construction Kit and SOCK with period-accurate engine options.

Adds RS-25 engine variants with deploy animations and Waterfall plumes. Engines use LH2/LOX via CRP. Designed to supplement Artemis CK and shuttle builds. CryoTanks recommended for managing LH2 boil-off on long missions.

Boring Crew Services

Dependencies: B9PartSwitch, ModuleManager

Impact: Parts

Version: Compatible with KSP 1.12.x. Pairs well with HabTech2 for ISS resupply and crew rotation missions.

Boeing Starliner (CST-100) crew capsule. Adds the Starliner command pod with integrated RCS, parachutes, airbags, and service module. Designed for Commercial Crew-style missions to LKO space stations.

Adds the Starliner crew capsule. Integrates with HabTech2 and StationPartsExpansionRedux for ISS-style station missions. Complements Artemis CK (SLS/Orion) and Tundra Exploration (Dragon) for a complete modern crew vehicle lineup.

HabTech2

Dependencies: B9PartSwitch, CommunityResourcePack, ModuleManager, Benjee10-SharedAssets, HabTechProps, HabTechRobotics, Waterfall

Impact: Parts, Mechanics

Version: Compatible with KSP 1.12.x. Requires Breaking Ground DLC. StationPartsExpansionRedux is strongly recommended (adds Russian segment and additional station modules). Boring Crew Services and Tundra Exploration provide crew vehicles.

International Space Station parts. Build the US Orbital Segment from modular truss, solar array, radiator, node, lab, and cupola parts. Includes Canadarm2, docking adapters, and functional solar array rotation. The definitive ISS construction mod.

Adds modular ISS parts with functional solar tracking, robotic arms, and docking systems. Requires Breaking Ground DLC for robotic Canadarm functionality. Integrates with StationPartsExpansionRedux for Russian segment and additional station modules.

Stockalike Station Parts Expansion Redux

Dependencies: ModuleManager

Impact: Parts

Version: Compatible with KSP 1.12.x. Integrates with HabTech2 for ISS builds. Community Tech Tree adds dedicated station-part tech nodes.

Comprehensive space station parts pack in stockalike style. Inflatable habitats, centrifuges, cupolas, trusses, cargo containers, docking hubs, and structural adapters — everything needed to build orbital stations from small outposts to sprawling complexes.

Adds stockalike station parts across multiple sizes and tech nodes. Parts are designed to complement stock station building while greatly expanding options. Works standalone or alongside HabTech2 for ISS-specific builds.

Mark One Laboratory Extensions (MOLE)

Dependencies: B9PartSwitch, ModuleManager

Impact: Parts, Mechanics

Version: Compatible with KSP 1.12.x. Designed for early career station-building. Pairs with StationPartsExpansionRedux and HabTech2 for a complete station-part progression from early Mk1 to late-game ISS-scale.

Early-game space station and orbital laboratory parts in the Mk1 form factor. Adds compact lab modules, experiment storage, science instruments, and orbital workshop parts that unlock early in the tech tree — before the stock Mobile Processing Lab. Build your first orbital outpost with 1.25m parts.

Adds Mk1-sized station and laboratory parts for early career mode. MOLE parts unlock before the stock lab, giving you a reason to build orbital stations early. Complements StationPartsExpansionRedux (later, larger station parts) and HabTech2 (ISS-specific).

Aircraft & Spaceplanes

Airplane Plus

Dependencies: ModuleManager

Impact: Parts

Version: Compatible with KSP 1.12.x. Parts are balanced for stock aerodynamics. Pairs well with Mk3 Expansion and Mark IV for a complete aircraft parts progression.

Adds propeller engines, helicopter rotors, vintage aircraft cockpits, and early-aviation structural parts. Extends atmospheric flight backward in the tech tree — build prop planes, biplanes, seaplanes, and helicopters before unlocking jet engines.

Adds propeller-driven aircraft parts across early tech nodes. Fills the gap between basic jets and the start of career mode. Helicopter rotors and vintage cockpits enable diverse atmospheric vehicle designs.

Mk3 Stockalike Expansion

Dependencies: ModuleManager

Impact: Parts

Version: Compatible with KSP 1.12.x. Mk2 Expansion (already included) handles Mk2-sized parts. Together they cover all stock fuselage profiles.

Expands the Mk3 fuselage part catalog: additional crew cabins, cargo bays, engine mounts, adapters, and structural parts. Triples the building options for Mk3-sized spaceplanes and heavy atmospheric craft.

Adds Mk3-profile parts at appropriate tech nodes. All parts maintain the stock Mk3 cross-section and attachment rules. Complements Mk2 Expansion (already in Wave 1) for the full fuselage-size progression.

Mark IV Spaceplane System

Dependencies: ModuleManager

Impact: Parts

Version: Compatible with KSP 1.12.x. Mk4 parts are late-game — you won't unlock them until you've mastered Mk2 and Mk3 designs.

Adds a Mk4 fuselage profile — larger than Mk3 — with dedicated crew cabins, cargo bays, fuel tanks, engine mounts, and aerodynamic nose/tail sections. For building massive SSTOs, heavy cargo spaceplanes, and interplanetary crew transports.

Adds Mk4-profile parts in the late tech tree. The Mk4 cross-section is significantly larger than Mk3 — these are the biggest spaceplane parts available. Designed for lifting heavy payloads to orbit in a single stage.

Mk-33

Dependencies: ModuleManager

Impact: Parts, Mechanics

Version: Compatible with KSP 1.12.x. KerbalKonstructs and DockRotate recommended. KIS/KAS, kOS, and life support mods are supported but optional.

Venture Star / X-33 inspired parts: lifting-body spaceplane with integrated aerospike engines, thermal protection tiles, and vertical-launch horizontal-landing profile. A single-stage-to-orbit spaceplane that launches vertically and glides back to the runway.

Adds a Venture Star-style spaceplane system with linear aerospike engines and metallic TPS. Designed for SSTO operations — launch vertically, reach orbit in a single stage, reenter and land horizontally. A unique design challenge distinct from conventional spaceplanes.

Early Game

Sounding Rockets

Dependencies: ModuleManager

Impact: Parts

Version: Compatible with KSP 1.12.x. Designed for Career mode — less useful in Science or Sandbox. Community Tech Tree places these in dedicated early nodes. CNAR (manual install) is a heavier alternative with more rocket variety.

Ultra-early-game rocket parts for Career mode. Tiny 0.35m sounding rockets, low-power solid motors, basic science instruments, and lightweight avionics. Lets you start launching suborbital science missions before unlocking the Stayputnik probe core — the true beginning of your space program.

Adds small-diameter rocket parts for the earliest stages of Career mode. Sounding rockets unlock before the first stock probe core, giving you something to launch in the first few science nodes. Parts are cheap, light, and perfect for "first launch" contracts.

IVA Enhancement

RasterPropMonitor

Dependencies: ModuleManager

Impact: Graphics, UI

Version: Compatible with KSP 1.12.x. DE_IVAExtension (see below) is recommended to add RPM-enabled IVAs to stock command pods. ASET Props and ASET Avionics (CKAN) add additional functional cockpit props.

Adds functional Multi-Function Displays (MFDs), instrument panels, and interactive screens to IVA cockpits and crew cabins. Transforms IVAs from static decoration into functional flight decks — view orbit info, docking cameras, and vessel data from inside the cockpit. Essential for IVA-only playthroughs.

Replaces static IVA props with interactive MFD screens. MFDs can display orbital data, docking cameras, radar, and custom instrument pages. Works with stock pods and most mod cockpits. DE_IVAExtension adds enhanced IVAs for specific pods using RPM.

DE_IVAExtension

Dependencies: RasterPropMonitor, ModuleManager

Impact: Graphics

Version: Compatible with KSP 1.12.x. Requires RasterPropMonitor. FreeIva (already in Wave 1) lets kerbals walk between DE-enhanced IVAs.

Enhanced IVA interiors for stock command pods. Replaces the default bare-bones IVAs with detailed, RPM-enabled cockpits featuring functional MFDs, switches, and instrument panels. Makes every stock pod and cockpit feel like a real spacecraft interior.

Replaces stock IVA models with detailed RPM-integrated interiors. Each pod gets a unique, functional cockpit layout. Requires RasterPropMonitor for the MFD functionality. FreeIva enables moving between these enhanced IVA spaces.

Procedural Parts

Dependencies: ModuleManager

Impact: Parts, Mechanics

Version: Compatible with KSP 1.12.x. Stock fixed tanks are the default. Install this only if you prefer design freedom over the intended engineering constraints of stock tanks.

Design decision — stock fixed tanks are the modlist default. Procedural Parts generates fuel tanks, structural fuselages, and SRBs in arbitrary shapes and sizes, reducing part count and giving unlimited design flexibility at the cost of stock balance and craft-sharing compatibility.

Replaces fixed-size parts with procedurally-generated versions. Tank shape, diameter, and length are adjustable via sliders. Part mass, cost, and capacity scale mathematically — balance differs from hand-tuned stock tanks. Significantly reduces part count on large vessels.

⚠ Conflicts: Stock fixed tanks are the recommended approach for this modlist. Procedural parts change the design puzzle and balance of rocket construction. Craft files using procedural tanks require the recipient to also have this mod installed.

Procedural Fairings

Dependencies: ModuleManager

Impact: Parts

Version: Compatible with KSP 1.12.x. Stock fairings are the default. Install only if you regularly build irregular payloads or interstage fairings that the stock editor can't handle.

Design decision — stock fairings are the modlist default. Procedural Fairings generates payload fairings of any size and shape with purpose-built interstage adapters and custom nose cones. Useful for irregularly-shaped payloads where stock fairings fall short.

Adds parts for generating procedural fairings and interstage adapters. Fairing shape is controlled by adjusting base/top diameters and height. More flexible than stock fairings for unusual payloads and interstage configurations.

⚠ Conflicts: Stock fairings are the recommended approach for this modlist — they cover 95% of use cases with a more intuitive editor. Procedural fairings add parts to the catalog and require the mod to be installed for craft file sharing.

Outer Planets Mod

Dependencies: ModuleManager, Kopernicus

Impact: Parts, Mechanics

Version: Compatible with KSP 1.12.x. Requires Kopernicus Stable Branch. Parallax support via Parallax-OuterPlanetsMod on CKAN.

Adds Saturn, Uranus, Neptune, and Pluto analogs to the stock Kerbol system — Sarnus, Urlum, Neidon, and Plock — each with their own moon systems. Seamlessly integrates beyond Jool's orbit with stock-quality terrain and science definitions.

Adds 4 new planets and 15+ moons to the outer Kerbol system. Each body has unique biomes, science, and terrain. Requires Kopernicus (the planet-pack framework). Delta-v requirements roughly triple for outer planet missions compared to Jool.

⚠ Conflicts: Kcalbeloh and OPM occupy the same region — installing both adds both systems but may make the outer solar system crowded. They are technically compatible but consider whether you want two competing outer systems.

Kcalbeloh System

Adds an interstellar system centered on a black hole (Kcalbeloh). Multiple planets orbit the singular-

Dependencies: ModuleManager, Kopernicus

Impact: Parts, Mechanics, Gameplay

Version: Compatible with KSP 1.12.x. Requires Kopernicus Stable Branch. Parallax config must be downloaded manually from the Kcalbeloh forum thread.

ity, including habitable worlds, gas giants, and exotic bodies. Accessible via wormhole or interstellar travel.

Adds an entire secondary star system reachable via a wormhole near Jool (or via very-long-distance interstellar transfer). Dramatically extends the endgame — reaching and exploring Kcalbeloh is a campaign-scale endeavor. Parallax terrain patch available from the mod's forum thread (manual install — not on CKAN).

⚠ Conflicts: Kcalbeloh and OPM both add bodies beyond Jool. They are technically compatible (Kcalbeloh is a separate star system with its own wormhole entrance) but running both is a very large install.

Minor Planets Expansion

Dependencies: Kopernicus, OuterPlanetsMod, KSPCommunityFixes, ModuleManager

Impact: Parts, Mechanics

Version: Compatible with KSP 1.12.x. Requires Outer Planets Mod and Kopernicus. Adds bodies without replacing or modifying OPM planets.

Adds dwarf planets, asteroids, and minor bodies to the Outer Planets Mod system. Populates the gaps between OPM's gas giants with realistically-scaled minor planets, trojan asteroids, and Kuiper belt objects. Requires OPM.

Adds minor bodies to the OPM system. Requires Outer Planets Mod. Each body has unique biomes and science definitions. Increases the number of destinations in the outer system without adding new star systems or altering existing planets.

QuackPack

Dependencies: Kopernicus, ModuleManager

Impact: Parts, Mechanics

Version: Compatible with KSP 1.12.x. Requires Kopernicus. Works alongside OPM without conflicts.

Inner system expansion by the Kopernicus team. Adds new planets and moons between Moho and Kerbin's orbit — fills the empty inner solar system with new destinations. Designed to complement OPM (outer system) for a fully expanded Kerbol system.

Adds new bodies to the inner Kerbol system. Requires Kopernicus. Complements OPM — OPM expands outward, QuackPack expands inward. Both can be installed together for a complete solar system expansion.

Manual Install Mods

These mods are not available through CKAN. Install them manually by downloading from their forum threads and placing the extracted folders in `GameData/`.

Manual-install mods do not appear in CKAN metapackages. You must download and update them yourself. Only install mods from mods you trust — always verify download links from the official forum threads.

Grannus Expansion Pack (GEP)

Dependencies: Kopernicus, ModuleManager

Impact: Parts, Mechanics, Gameplay

Version: Compatible with KSP 1.12.x. Manual install only. Requires Kopernicus.

Adds a binary star system centered on Grannus, a red dwarf companion to Kerbol. Multiple planets orbit the distant star, reachable via interstellar transfer. Expands the late-game with a second star system. GEP-Volumetrics (SpaceDock) adds cloud support.

Adds the Grannus binary star system. Not on CKAN — manual install from forum thread. Compatible with OPM and Minor Planets Expansion.

⚠ Conflicts: Kcalbeloh also adds a secondary star system. Both can be installed together for two interstellar destinations, but this significantly increases memory usage. Consider whether you want one or both secondary systems.

OPT Spaceplane Continued

Dependencies: B9PartSwitch, ModuleManager

Impact: Parts

Version: Compatible with KSP 1.12.x. Manual install only. OPT Reconfig (CKAN) required for Community Tech Tree integration and balance adjustments.

Futuristic spaceplane parts with unique sci-fi profiles, advanced hybrid engines, and large-diameter cargo bays. The continued fork of the original OPT (Orbit Portal Technology) mod, maintained for KSP 1.12.x. For players who want spaceplanes beyond the stock/procedural aesthetic.

Adds futuristic spaceplane parts with advanced engine types. Not on CKAN — manual install from forum thread. OPT Reconfig (on CKAN) is a companion config mod that adjusts OPT balance and adds CTT support.

Completely Non-Aggressive Rocketry (CNAR)

Dependencies: ModuleManager

Impact: Parts

Version: Compatible with KSP 1.12.x. Manual install only. Sounding Rockets (CKAN) is

Expanded rocket parts pack with unique engine configurations, fuel tank geometries, and structural components. A parts-heavy rocket-building mod for players who want more variety in their launcher designs beyond stock, procedural, and historical mods. Includes LFO, monoprop, and solid rocket components.

Adds a broad collection of rocket parts. Not on CKAN — manual install from forum thread or SpaceDock (id:2271).

the recommended early-game rocket mod for this modlist. CNAR adds more variety but requires manual installation.

A heavier alternative to Sounding Rockets for expanding the early-to-mid game rocket catalog.

KerbalFX

Dependencies: ModuleManager

Impact: Graphics

Version: Compatible with KSP 1.12.x. Manual install from SpaceDock — not on CKAN. Works alongside Firefly and VaporCones for a complete aero-effects suite.

Additional aerodynamic effects: improved contrails, wingtip vortices, condensation effects, and enhanced vapor cones. Very light on performance — adds visual polish to atmospheric flight without the GPU cost of heavy shader mods. Complements Firefly and VaporCones with more subtle atmospheric particle effects.

Adds aerodynamic particle effects during atmospheric flight. Lightweight — designed to run well even on modest GPUs. No gameplay impact.

Kerbin Side Remastered

Dependencies: KerbalKonstructs

Impact: Parts

Version: Compatible with KSP 1.12.x. Manual install from SpaceDock — not on CKAN. Requires KerbalKonstructs (available via CKAN suggests).

Adds dozens of detailed bases, airfields, and points of interest across Kerbin. Landing strips, helipads, radar installations, and even an underground hangar — makes Kerbin feel alive and worth exploring. Each location has unique features and serves as a destination for atmospheric flight missions.

Adds static structures and launch sites via KerbalKonstructs. No flight mechanics changes — purely adds locations to visit. Expands the scope of atmospheric flight from “go up and come down” to actual cross-Kerbin navigation between points of interest.

Throttle Controlled Avionics

Dependencies: ModuleManager

Impact: UI, Mechanics

Version: Compatible with KSP 1.12.x. Manual install from SpaceDock — not on CKAN. Essential for practical VTOL operations. Configure engine

VTOL flight computer. Automatically balances thrust across multiple engines to maintain stable hover and controlled translation — essential for VTOL craft, tilt-engine designs, and precision landers. Transforms VTOL from a white-knuckle balancing act to a flyable, practical vehicle class.

Adds thrust-balancing and VTOL control algorithms. Manages differential throttle across engine groups to maintain attitude during hover and translation. No new parts — works with any engine configuration. Complements At-

groups in the VAB for TCA to manage.	mosphereAutopilot and MechJeb with VTOL-specific control.

Wave 1 Mods — How to Use Them

MechJeb

MechJeb is an autopilot with a learning curve. Every feature is accessed through the MechJeb menu (toolbar button or the part you attach to your craft). The window has tabs: Ascent Guidance, Landing Guidance, Rendezvous Autopilot, Maneuver Planner, and more.

How to learn from MechJeb:
1. Manually execute a maneuver first — say, a Mun transfer burn
2. Watch MechJeb do the same burn next time
3. Note the burn timing, throttle profile, and attitude corrections
4. Eventually, you'll internalize the technique and can do it without the autopilot

Ascent Guidance: Set your target orbit (altitude, inclination) and click "Engage Autopilot." MechJeb handles the gravity turn, staging, and circularization. Useful for routine satellite launches when you've done the same ascent a hundred times.

Maneuver Planner: Create complex multi-step maneuvers (transfer to planet, match planes, circularize) as a sequence. MechJeb executes them in order. This is powerful for Duna and Jool missions.

Landing Guidance: Pick a target on the surface. MechJeb deorbits and lands within a few hundred meters. Great for precision base-building in Wave 2.

> MechJeb's data readouts are similar to KER's. If you prefer MechJeb's autopilot but KER's cleaner HUD, you can use both — open MechJeb for automation, KER for the flight HUD.

AtmosphereAutopilot

AtmosphereAutopilot (AA) is a fly-by-wire system for atmospheric flight. It's more sophisticated than SAS for aircraft.

- **Auto-trim:** Automatically adjusts control surfaces so your plane flies straight hands-off. Toggle on before long cruise flights.
- **Altitude hold:** Maintains current altitude using pitch control. Good for survey contracts at constant altitude.
- **Heading hold:** Maintains current heading. Combines with altitude hold for hands-off cruise.
- **Auto-throttle:** Holds a target speed. Use for efficient subsonic cruise or supersonic dashes.

- **Fly-by-wire modes:** The “Moderation” mode smooths your inputs, preventing over-correction. “Director” mode flies toward your cursor position. Toggle between them in AA’s settings.

> AA handles atmosphere — MechJeb handles space. Use both. AA’s altitude/heading hold for the atmospheric climb, then hand off to MJ’s ascent autopilot at 30 km. They don’t conflict.

Kerbal Atomics + System Heat

Nuclear engines and reactors produce heat. System Heat is the thermal management framework — every reactor and NERV variant in Kerbal Atomics uses it.

Reactor Basics

1. **Startup:** Right-click the reactor, click “Activate.” It takes time to reach operating temperature — nuclear engines produce zero thrust while warming up (30–120 seconds depending on reactor size).
2. **Shutdown:** Right-click, “Deactivate.” The reactor cools passively but stays hot for hours. You cannot time-warp through cooldown.
3. **Waste heat:** Heat is stored in the reactor and radiators. If the “Waste Heat” bar fills completely, the reactor emergency-scrambles and you lose all power/thrust.

Radiator Rules

- Radiators must be deployed (extendable panels) OR active (fixed panels that glow when cooling)
- One medium radiator per small reactor; 4–8 large radiators per fusion reactor
- Radiators cool everything on the same vessel — you don’t need to pair them
- Place radiators symmetrically and away from other hot parts

> The System Heat toolbar button shows a live thermal overlay. Green = nominal, yellow = warming, red = approaching shutdown. If everything is red, deploy more radiators or reduce power.

Using the Graphics Mods

Wave 1 adds a full visual overhaul stack. Here’s how to configure each layer:

- **Scatterer:** Atmospheric scattering, ocean shaders, sunflare. Toggle in flight via the Scatterer toolbar menu. Adjust ocean quality if you see FPS drops during launch.

- **EVE + AVP:** Cloud layers and skybox. Installed automatically. No manual config needed unless you want to tweak cloud density.
- **Deferred:** Changes the entire render pipeline. Toggle via config file in GameData/Deferred/. If you see visual glitches, disable it — the game falls back to stock rendering. Deferred handles planet reflections, so no separate planet-lighting mod is needed.
- **TUFX + Shabby/Shaddy:** Post-processing (bloom, ambient occlusion, color grading). Switch profiles from the TUFX toolbar menu in flight. Shabby and Shaddy add extra profile presets beyond the built-in defaults. Try Shabby first for a cinematic look.
- **Parallax:** Terrain tessellation. In Parallax settings (toolbar button), reduce scatter density if you see terrain pop-in or FPS drops near the surface. High-quality setting requires a DX11-capable GPU.
- **Waterfall + Restock Waterfall Expansion:** Engine plumes. Automatic — no config needed. The plumes respond to atmospheric pressure (expand in vacuum, contract in atmosphere). Purely visual.
- **Rocket Sound Enhancement:** Audio overhaul. Sonic boom effects, distance-based attenuation. Configurable in settings — disable sonic booms if they startle you during timewarp transitions.
- **Distant Object Enhancement:** Renders distant planets and vessels as points of light. The toolbar menu lets you toggle vessel flares and adjust sky-dimming intensity.
- **Textures Unlimited + TURD + Simple Repaint:** PBR shading framework and part recoloring. TURD adds a repaint GUI in the VAB/SPH — right-click any supported part to change its color. Simple Repaint covers parts without TURD configs. Both are visual only.

New Parts Overview

Wave 1 adds a lot of parts. Here's how they're organized:

- **Restock + Restock+:** Revamped stock parts (visual) + new stock-alike parts (functional). Your rockets look better and you get missing part sizes like 0.625m RCS tanks and 2.5m reaction wheels.
- **Near Future suite (7 packs):** The core parts expansion. Solar panels (blanket arrays for outer system), electrical (reactors + capacitors), propulsion (advanced engines), construction (trusses + structural), spacecraft (command pods), launch vehicles (5m–7.5m lifters), and aeronautics (spaceplane parts). NF Launch Vehicles covers the heavy-lift niche; NF Electrical's reactors power the entire tech tree.
- **Historical:** Bluedog Design Bureau (US rockets from Mercury to Apollo) and Tantares (Soviet spacecraft + launchers). Both are comprehensive — each adds 100+ parts. Pick one or install both for the full Cold War experience.
- **Modern:** Artemis Construction Kit (SLS + Orion), Eisenhower Astronautics (modern launchers), Tundra Exploration (SpaceX-inspired). Each covers a different modern program.

- **Station parts:** Stockalike Station Parts Expansion Redux, HabTech2, Planetside Exploration Technologies. Everything you need for orbital construction — inflatable habitats, centrifuges, trusses, docking adapters.
- **Planes:** Airplane Plus, Mk2/Mk3/MkIV expansion packs, Mk-33, OPT Spaceplane. If it flies in atmosphere, it has expanded parts here.
- **Planet packs:** Outer Planets Mod (Saturn–Pluto analogs), Minor Planets Expansion (dwarf planets between them), QuackPack (inner system), Kcalbeloh (black hole system), Grannus (binary red dwarf). Each adds new destinations with full biome and science support. You can install all five if your system handles it.

The Fuel System

Wave 1 introduces custom fuel types beyond stock LiquidFuel/Oxidizer. Three mods power this ecosystem.

Community Resource Pack (CRP)

CRP defines the resources: Hydrogen, Methane, Kerosene, Hydrazine, and dozens more. You never interact with CRP directly — it's a library that other mods reference. But without it, custom fuels don't exist.

B9 Part Switch

B9PartSwitch is the part you actually use. In the VAB, right-click any compatible fuel tank. A dropdown lets you switch what it holds: LFO (stock), LH2/Oxidizer (cryogenic), Methane/Ox, or Monopropellant. The tank model doesn't change — only its contents.

> LH2 is the most efficient fuel (highest Isp) but the least dense. A tank full of LH2 weighs less than the same tank full of LFO — but it takes up more volume. LH2 stages are physically larger for the same delta-v.

Cryogenic Tanks — Boil-Off Management

LH2 and Methane are cryogenic — they slowly evaporate in storage. This is "boil-off."

- **Standard tanks:** Fuel evaporates over time. Acceptable for short missions (Mun, Minmus). Unacceptable for interplanetary.
- **Insulated tanks:** CryoTanks adds insulated variants that reduce boil-off by 90%. Use these for Duna and beyond.
- **Active cooling:** Some parts (cryo-coolers) consume ElectricCharge to eliminate boil-off entirely. Necessary for Jool missions lasting years.

> Boil-off only matters on long time-scales. A Mun mission takes days — you'll never notice. A Grannus transit takes decades — uninsulated LH2 will be gone before you leave Kerbin's SOI.

Planet Packs: OPM and Kcalbeloh

Outer Planets Mod (OPM): Adds 4 planets (Sarnus, Urlum, Neidon, Plock) beyond Jool's orbit, each with a full moon system. Delta-v costs roughly triple for outer planets compared to Jool, and travel times are measured in **years**. OPM integrates seamlessly — stock contracts will generate for these bodies, and science experiments have unique biome definitions.

Kcalbeloh System: An entirely separate star system accessible via a wormhole near Jool. The wormhole behaves like a special SOI — fly into it and you emerge in the Kcalbeloh system. The delta-v to reach the wormhole is roughly equivalent to a Jool transfer (2,000 m/s from LKO). Once inside Kcalbeloh, the system's planets range from habitable worlds to exotic bodies orbiting a black hole.

> Installing both OPM and Kcalbeloh makes for a very large (and somewhat crowded) outer system. OPM fills the gap between Jool and interstellar space; Kcalbeloh adds a separate star system. They are technically compatible, but you may prefer to choose one per save.

Parallax terrain patches:

- OPM: Install `Parallax-OuterPlanetsMod` from CKAN. This automatically applies Parallax terrain to OPM bodies.
- Kcalbeloh: The Parallax patch is NOT on CKAN. Download it from the Kcalbeloh forum thread and place it in GameData manually.

Performance Tuning

Wave 1 adds significant performance load, especially from graphics mods. If your game runs poorly:

1. **Disable TUFX ambient occlusion** — this alone can recover 10–20 FPS
2. **Reduce Parallax scatter density** to 50%
3. **Lower AVP texture resolution** or remove it entirely
4. **Cap part counts** — keep ships under 200 parts, stations under 300

> KSP is CPU-bound by physics, not GPU-bound by graphics. If you have FPS issues during launch (when physics is heavy) but not in orbit, the culprit is part count, not your graphics mods. Simplify the vessel, not the visuals.

Wave 1 Guide — Going Further

Build Strategies for Interplanetary Ships

Interplanetary ships are different from launch vehicles. You build them in orbit and they never touch atmosphere. Design accordingly.

Staging in Vacuum

In space, there's no drag and no gravity losses. You don't need high TWR. What matters is **delta-v per ton of dry mass**.

- **Serial staging:** Drop empty tanks as you go. Simple, reliable, wastes docking ports.
- **Asparagus staging:** Feed fuel inward from outer tanks. All engines fire at once. Best mass fraction, complex plumbing.
- **Nuclear transfer stages:** A single NERV engine on a long Mk3 fuselage of liquid fuel only (no oxidizer). 8,000 m/s in a single stage. The interplanetary workhorse.
- **Ion tugs:** Dawn engines on xenon. 20,000+ m/s, but burns take hours and thrust is measured in millinewtons. Good for small probes, terrible for crewed ships.

> After 3,000 m/s of delta-v in a single stage, adding more fuel tanks gives diminishing returns. The tank's own dry mass eats your gains. At that point, add another stage — or switch to a higher-Isp engine.

TWR vs Isp — What Matters in Space

- **Launch:** TWR > 1.3. Isp is secondary. You're fighting gravity.
- **Transfer burn:** TWR > 0.3. Lower means the burn takes multiple orbits (periapsis kicking). Annoying but doable.
- **Deep space cruise:** TWR 0.05 is fine. You have months. Isp is everything.
- **Landing:** TWR > 1.0 relative to the body you're landing on (Mun: 1.6 m/s², so 0.2 TWR relative to Kerbin). Lightweight vacuum engines like the Terrier or Poodle excel here.

Modular Ship Design

Build interplanetary ships as dockable modules:

1. **Propulsion section** — engine cluster + fuel tanks. Detachable and reusable.

2. **Payload** — lander, rover, station module. Dock to the tug for transit, undock at destination.
3. **Crew module** — hitchhiker container or Mk2 crew cabin with docking port.
4. **Power + comms** — solar panels or RTG, relay antenna, probe core (always).

> The modular approach means your deep-space tug does multiple missions. After delivering a Duna lander, the tug returns to Kerbin orbit, refuels at a depot, picks up a Jool payload, and goes again. Reusable infrastructure saves launches — and funds in Career mode.

The Delta-V Budget

Before launching a mission, add up the cost of every maneuver:

- LKO to Duna transfer: 1,100 m/s
- Duna capture (aerobrake): 0 m/s
- Land on Duna (parachutes): 50 m/s
- Duna ascent to orbit: 1,400 m/s
- Duna to Kerbin transfer: 700 m/s

Total one-way: 3,250 m/s. Round-trip (no refuel): 6,500 m/s.

Design your ship to the round-trip number first, then add 20% margin for mistakes. If the resulting ship is absurdly large, consider ISRU refueling at the destination.

Campaign Play — Career Mode Strategy

Wave 1 mods transform Career mode. The Community Tech Tree deepens progression from 15 nodes to 50. Near Future parts fill those nodes with meaningful upgrades. Here's how to play it.

Contract Selection

Not all contracts are worth your time.

- **Accept:** “Explore [body]” (big payout, drives progression), “Science from [body]” (pairs with exploration), satellite contracts (easy money with a relay bus).
- **Skip:** Part-testing contracts (tedious, low pay), rescue contracts (fun but scale poorly), tourist contracts (acceptable early, tedious late).
- **Must-take:** World First milestones. These are the game’s progression backbone and pay enormously.

> Build a “contract bus” — a small probe with every science instrument, a relay antenna, and 3,000 m/s of delta-v. Launch one to each new body you visit. It completes 3–4 contracts at once.

Science Farming

With CTT installed, you need roughly 3× the science to complete the tree. Priority order:
1. **Mun + Minmus biome hopping** — a single lander with all experiments can visit 5+ biomes per trip. Bring a scientist to reset the Science Jr. and Goo.
2. **Mobile Processing Lab** — put one in Minmus orbit, feed it data from the surface, collect 500 science per transmission.
3. **SCANsat** — scanning planets generates science passively. Launch scanner satellites to every body in the system.
4. **Kcalbeloh/Grannus** — each interstellar body is a fresh science goldmine.

Funds Management

- Upgrade Mission Control first (more contracts), then Tracking Station (patched conics for interplanetary), then R&D (unlock higher tech nodes).
- The VAB and Launchpad upgrades are expensive — wait until you have 1M+ funds.
- Tourism contracts to the Mun and Minmus are the best funds/hour in the mid-game. A 16-seat tourist bus to Minmus pays 500,000 funds.

CTT Node Priority

With Community Tech Tree, the stock “one node unlocks everything” problem is fixed. Each mod’s parts sit in dedicated nodes. Priority path for Wave 1:

1. **Basic Science → Space Exploration:** Unlock SCANsat parts and basic probes
2. **Nuclear Propulsion:** NERV and Kerbal Atomics engines — your interplanetary workhorses
3. **Large Probes → Advanced Electrics:** Near Future Solar panels. The blanket arrays are game-changers for outer-system missions.
4. **Advanced Fuel Systems → Cryogenic Engines:** CryoTanks and LH2 engines for high-Isp upper stages
5. **Orbital Assembly → Large Station Parts:** StationPartsExpansionRedux — build your orbital fuel depot

After this, specialize: colonization (Planetside, MKS), interstellar (FFT, Blueshift), or expand (more planet packs).

Interplanetary Transfer Windows

Phase Angles

Every planet has a specific launch window when the transfer is most efficient. The **phase angle** is the angle between your origin planet, the Sun, and the destination planet. Key windows:

- **Duna** (Mars analog): Phase angle 44°, delta-v from LKO 1,100 m/s
- **Eve** (Venus analog): Phase angle 54°, delta-v from LKO 1,100 m/s
- **Jool** (Jupiter analog): Phase angle 96°, delta-v from LKO 2,000 m/s
- **Moho** (Mercury analog): Phase angle −252°, delta-v from LKO 2,200 m/s

> Duna is the best first interplanetary target. It has an atmosphere (thin, but enough to aerobrake and use parachutes), low gravity, and a moderate transfer cost. Eve is easier to reach but brutally hard to leave — its thick atmosphere and high gravity make ascent nearly as expensive as Kerbin.

Executing a Duna Transfer

1. Wait for the Duna transfer window (phase angle 44°). You can eyeball it: Duna should be about 1/8 of an orbit ahead of Kerbin.
2. From LKO, set Duna as your target.
3. Create a maneuver node. Pull prograde until the projected orbit touches Duna's. Adjust the node position until you get an encounter.
4. Burn. Mid-course correction: about halfway there, create another tiny node to fine-tune your Duna periapsis.
5. At Duna, aerobrake — set periapsis to 15–20 km to use the atmosphere for capture. Bring heat shields.

> Ike (Duna's moon) is tidally locked and small — an even easier landing target than the Mun. If Duna itself intimidates you, go to Ike first.

> **You're not in Kerbin's orbit anymore.**
>
> Cross the sphere-of-influence boundary and something changes. The navball still works. The engines still fire. But the blue planet that has been your home for every mission until now just became a dot. Duna is ahead — red, alien, waiting. Between you and it: millions of kilometers of nothing.
>
> Every interplanetary mission begins with this moment. The point where Kerbin stops being your world and becomes a reference point. You're not going home for a while. You're going somewhere new.

Eve – The Purple Hell

Eve is the easiest planet to reach and the hardest planet to leave. This is the game's ultimate engineering challenge.

Transfer and Entry

- Transfer: Phase angle 54°, delta-v from LKO 1,100 m/s (same as Duna — deceptively cheap)
- Atmospheric entry: Eve's atmosphere is 5× denser than Kerbin's. You will need heat shields — and you won't need engines until you leave.
- Landing: Parachutes work extremely well. A single Mk16-XL can land 20+ tons. No engines needed for touchdown.

Surface Operations

Eve's surface is hot, purple, and high-pressure. Solar panels work fine (no atmosphere attenuation issues), but your kerbals are stuck unless you brought a serious ascent vehicle.

- **Science:** Eve has biomes (peaks, lowlands, shallows, etc.) but they're hard to reach without an aircraft.
- **Explodium Sea:** Liquid on Eve's surface. Ships float (barely). Not recommended for first visits.

> Do not send a crewed mission to Eve unless you have a tested, working ascent vehicle. Eve ascent from sea level costs 8,000 m/s of delta-v — nearly three times Kerbin. The atmospheric pressure kills engine Isp until 30 km altitude. This is the single hardest maneuver in stock KSP.

Eve Ascent Strategy

1. **Do not land at sea level.** Target a mountain peak (5+ km altitude). Every kilometer of elevation saves 500 m/s.
2. **Aerospike engines** (Dart) maintain decent Isp in thick atmosphere. Vector engines work for the upper stage.
3. **Shed everything.** Jettison parachutes, landing legs, ladders — all dead weight — before lighting engines.
4. **Stage aggressively.** Your first stage gets you through the soup (0–20 km). Second stage takes over when Isp recovers.
5. **Fairing or nose cone** on top. Drag is brutal in Eve's lower atmosphere.

Gilly — Eve's Lifeboat

Gilly is Eve's tiny captured asteroid-moon. Gravity: 0.005 g. You can reach orbit with RCS alone.

- **Why visit:** Gilly has high ore concentration and trivial escape costs. It's the best ISRU base in the inner solar system.
- **How to land:** You don't "land" on Gilly — you rendezvous with it. Approach at < 5 m/s. Time warp kills relative velocity.
- **Strategy:** Build a Gilly mining outpost. Refuel Eve ascent vehicles in Gilly orbit before descending. This solves the "how do I get back from Eve" problem without needing a single-stage-to-orbit-from-sea-level monster.

Moho — The Sun-Diver

Moho is Kerbin's closest planet to the sun — and one of the hardest to reach despite being "right there."

Why Moho is Hard

- **Inclined orbit:** Moho's orbit is tilted 7° relative to Kerbin's. You must match inclination mid-transfer or at arrival — expensive either way.
- **Deep gravity well:** Transfer from LKO costs 2,200 m/s. Capture at Moho costs another 2,000 m/s. Total one-way: 4,200 m/s — more than Jool.
- **No atmosphere:** Cannot aerobrake. Every meter per second must come from your engines.
- **No moons:** No ISRU helper body. You must bring all return fuel or mine on Moho's surface.

Transfer Window

Phase angle −252° (Moho is ahead of Kerbin in its orbit). This window is short — a few days at best. Use Transfer Window Planner to nail it. The ejection burn from Kerbin should also include a normal component to match Moho's inclination. TWP gives you the exact numbers.

Capture and Landing

Moho has no atmosphere and moderate gravity (0.27 g). Landing costs 1,200 m/s. Use a high-Isp vacuum engine (Terrier, Poodle, or a nuclear stage). The surface is hot but solar panels produce enormous power this close to the sun — small panels are enough.

> Moho has a large molten core and a thin crust. The "Mohole" at its north pole is a bottomless pit. Do not drive rovers near it. Kerbals have fallen in. They do not come out.

ISRU — Mining and Refueling

The ISRU Chain

In-Situ Resource Utilization converts raw ore into usable fuel. The components:

1. **Surface Scanner** — maps ore concentration from polar orbit
2. **Drill (e.g. Drill-O-Matic)** — extracts ore from the surface
3. **Ore Tank** — stores raw ore
4. **ISRU Converter (e.g. Convert-O-Tron)** — converts ore + electricity into liquid fuel, oxidizer, or monopropellant
5. **Radiators** — the converter generates immense heat and will shut down without sufficient cooling

> ISRU equipment is godlessly heavy. A full mining rig can weigh 20+ tons. Test the full chain on Minmus first — its low gravity makes landing and returning with heavy payloads far easier than the Mun. Minmus also has high ore concentrations in its flats.

Minmus Mining Base — A Walkthrough

Minmus is the ideal ISRU starting point. Low gravity (0.05 g), flat landing zones, and high ore concentration in the Flats biome. Here's how to set up a fuel production chain.

Site Selection

- **Greater Flats:** Huge, perfectly level area near the equator. Easiest landing zone in the game. Ore concentration 8–12% (excellent).
- **Lesser Flats:** Smaller, slightly inclined. Backup option.
- **Poles:** Some ore, but inclined — harder to land and ascend efficiently.

Land your scanner satellite in polar Minmus orbit first (SCANsat or stock M700). The SCANsat map will show you exact ore hotspots.

Miner Design

A Minmus miner needs:

1. **Drill-O-Matic Junior** (or Senior for speed) — deployed with the “Deploy Drill” action
2. **ISRU Convert-O-Tron 125** (the small one is enough for Minmus)
3. **Ore tank** — at least the 300-unit radial tank
4. **Fuel tanks** — the miner is its own first customer. Fill them up on the surface.

5. **Power:** 4× Gigantor solar arrays + 2× Z-4K batteries. Minmus has good sunlight.
6. **Radiators:** 2× medium TCS panels. The converter melts without them.
7. **Engine:** A single Terrier or Spark. TWR > 0.3 on Minmus is trivial.

> The mining drill must touch the ground. Mount it low on the lander or use pistons (Breaking Ground DLC). If the drill hovers above the surface, it won't work — and the game won't tell you why.

The Fuel Tanker

Build a separate craft for ferrying fuel to orbit:

1. Large fuel tanks (at least a Rockomax X200-32)
2. A docking port (Jr. or standard) on top
3. RCS thrusters for docking at the orbital depot
4. Enough TWR to lift a full fuel load from Minmus surface (0.1 Kerbin TWR is plenty)

The tanker lands at the mining base, docks (or uses KAS/KIS fuel hoses), fills up, and returns to orbit. Each round trip costs 400 m/s. A full X200-32 tank of LFO is a substantial amount of fuel for interplanetary operations.

Orbital Fuel Depot

In low Minmus orbit (20 km), park a fuel depot:

1. Rockomax Jumbo-64 tank (the orange one) with docking ports on both ends
2. Large reaction wheels (full tanks are heavy and sluggish)
3. Relay antenna — the depot doubles as a comms relay for far-side landers
4. Probe core — no crew needed for a gas station

The workflow: Miner fills itself → tanker lands, docks, transfers fuel → tanker ascends, docks with depot → depot refuels interplanetary ships. This chain turns Minmus into infinite free fuel in Kerbin's backyard.

> Automate with Kerbal Alarm Clock: set an alarm for when the miner's ore tanks are full (6 days for a Junior drill with 8% ore). When the alarm fires, switch to the miner, launch the tanker, make the fuel run. One fuel run per week keeps your interplanetary fleet running indefinitely.

Duna Surface Operations

You've reached Duna. Now what? Surface ops turn a flag-planting mission into a sustained presence.

Landing Zones

- **Lowlands:** Flat, low elevation, rich in science. Safe landing zone for first missions.
- **Midlands:** Rolling terrain. More biomes nearby for rover exploration.
- **Highlands/Peaks:** Hard to land on, high science value. Good for experienced missions.
- **Poles:** Ice caps. Cold, inclined, but unique science.

Rover Deployment

Duna's gravity (0.3 g) and thin atmosphere make rovers viable but tricky:

- Wide wheelbase prevents flipping on slopes
- Low center of mass, reaction wheels set to "SAS only"
- Drive at < 15 m/s — Duna's surface looks smooth from orbit but is littered with rocks
- Solar panels work at 40% efficiency compared to Kerbin. Bring RTGs for long-duration rovers

Surface Base Design

A Duna base needs:

1. **Power:** Solar arrays + fuel cells for dust storms (yes, dust storms exist with visual mods; no, they don't block solar panels — but they look cool)
2. **ISRU:** Duna has ore. Mine it. The low gravity makes tanker ascents cheap.
3. **Habitation:** Hitchhiker containers or Planetside inflatables
4. **Docking:** Surface docking is hard — ports rarely align on uneven ground. Use KAS/KIS to connect modules with flexible pipes, or land modules on wheels and dock them on a flat area.

Ike — Duna's Relay Hub

Ike is tidally locked to Duna. A satellite in Ike-stationary orbit above Duna's far side has permanent line-of-sight to Ike AND Kerbin. Put a powerful relay there (RA-100) and every probe on Duna's surface has comms coverage through Ike.

> Ike has no atmosphere and low gravity (0.1 g). It's an even easier mining target than Minmus. Build your Duna-system fuel depot on Ike, not Duna. The savings in tanker ascent delta-v pays for the transfer from Ike orbit to Duna orbit many times over.

The Jool System

Jool is the Kerbol system's gas giant — 5 moons, each a unique challenge. With OPM installed, Jool is the gateway to the outer planets.

Transfer

Phase angle 96°, delta-v from LKO 2,000 m/s. Jool's massive gravity well makes capture easy — a small retro-burn at periapsis (200 m/s) captures you. The real cost is maneuvering between moons.

Moon Hopping Strategy

Visit in this order to minimize delta-v:

1. **Laythe (outermost):** Oxygen atmosphere. Jet engines work here. Land a spaceplane on the islands, explore the oceans. Joolrise is gorgeous.
2. **Vall (inner):** Europa analog. Ice moon, moderate gravity (0.23 g). Blue-tinted terrain. Good ISRU candidate — abundant ore.
3. **Tylo (deep gravity well):** The monster. Tylo has Kerbin-like gravity (0.8 g) and no atmosphere. Landing costs 2,500 m/s — more than reaching orbit from Kerbin. This is the ultimate vacuum landing challenge. Use high-TWR engines, bring extra fuel, and save before descent.
4. **Bop (outer, inclined):** Small captured asteroid. Low gravity, inclined orbit. Good mining base once you're in the Jool system.
5. **Pol (outermost):** Tiny, lumpy, low gravity. Tricky to land on (the terrain is spiky), but the views of Jool are spectacular.

Laythe Spaceplanes

Laythe is the only body beyond Kerbin with a breathable atmosphere. Jet engines (RAPIERs in air-breathing mode, or standalone Whiplash/Panther) work perfectly. A Laythe SSTO can fly indefinitely on atmospheric oxygen.

- **Island hopping:** Laythe's land is scattered islands. A seaplane or amphibious lander is essential.
- **Science:** Laythe has diverse biomes (shores, shallows, poles, islands). A single spaceplane can visit them all.
- **ISRU:** Ore is available, but you're competing with Vall for mining efficiency. Vall is better for fuel, Laythe is better for crew.

> **Jool is rising.**
>
> You are standing on an island on Laythe. Liquid water laps at the shore. Oxygen fills the air — jet engines work here, impossibly. And on the horizon, Jool fills half the sky. Green bands of cloud swirl in slow motion. Three moons are visible as crescents. The light is green-tinted, alien, beautiful.

> You flew here. Across the solar system. To an ocean moon. To watch a gas giant rise.
>
> There are people on Earth who have never seen anything this beautiful. You are looking at it through the window of a spaceplane you designed.

Tylo — The Final Exam

Tylo is what separates good engineers from great ones. Requirements:

- TWR > 1.0 on Tylo (0.8 Kerbin TWR)
- Delta-v from low Tylo orbit to surface and back: 5,000 m/s
- No atmosphere = no parachutes, no aerobraking. Pure engines.

> The optimal Tylo lander uses asparagus-staged drop tanks. Start with 4–6 radial tanks feeding a central core. Drop pairs as they empty. The final core stage has enough TWR to land alone (now much lighter). This is one of the few places where asparagus staging is unquestionably correct.

Bop and Pol — Mining Outposts

After conquering Tylo, Bop and Pol are your reward. Both have:

- Extremely low gravity (RCS landing viable)
- High ore concentrations
- Excellent ISRU base potential for outer-system missions (OPM planets beyond Jool)

> The Jool system is the last stop before the outer planets (OPM) and the first interstellar hop (Kcalbeloh wormhole, if installed). Build your Jool fuel depot well — it will service missions for the entire endgame.

> **The wormhole waits.**
>
> Near Jool, there is a distortion in spacetime — a sphere of bent light that leads to another star system. You can see the stars on the other side, shifted blue and strange. A black hole lives there. Planets orbit it. The laws of physics stretch thin around its event horizon.
>
> You don't have to go through. Nobody is making you. But you came all this way. And on the other side, there is something no kerbal has ever seen.
>
> The wormhole doesn't care if you're ready. It just is.

SSTO Spaceplanes

Design Principles

Single Stage To Orbit spaceplanes use jet engines to climb through the atmosphere, then switch to closed-cycle rocket mode for the final push to orbit. Key design considerations:

- **RAPIER engines** are the gold standard — they auto-switch between air-breathing and rocket mode
- **Center of mass vs. center of lift** — CoL must be behind CoM at all fuel levels. Check with tanks both full and empty in the SPH.
- **Wing area** — more is better. You need lift at high altitude where the air is thin.
- **Intake spam is dead** — KSP 1.0+ aerodynamics fixed this. One shock cone intake per 4 RAPIERs is plenty.

Ascent Profile

1. Accelerate along the runway to 140 m/s, pull up at 10–15 degrees
2. Climb at 15–20 degrees until 10,000 m
3. Level off to 5–10 degrees. Accelerate to 1,400 m/s at 20,000 m. This is where RAPIERs produce peak thrust.
4. When thrust drops (around 23–25 km), RAPIERs auto-switch to closed-cycle. Pitch up to 20–30 degrees.
5. Burn to apoapsis > 70 km, circularize as normal.

Relay Networks

The CommNet System

KSP's CommNet requires line-of-sight to Kerbin for probe control. Signal strength depends on antenna power, distance, and the tracking station level. A relay network places satellites between Kerbin and your destination to bounce the signal.

Relay Satellite Design

- Every relay needs a relay-capable antenna (RA-2, RA-15, RA-100)
- Pair with a direct antenna for the satellite's own connection back to Kerbin
- Include solar panels, batteries, a probe core, and reaction wheels
- Always add a small engine + fuel for final orbit adjustments

Kerbin Relay Constellation

For continuous coverage around Kerbin, launch 3–4 relay satellites equally spaced in a high circular orbit (2,800 km for 4-sat coverage). Launch them all on one rocket, detach at apoapsis, and circularize each individually.

Asteroid Capture

Finding Asteroids

Asteroids spawn near Kerbin and are tracked in the Tracking Station. Unknown objects must be discovered first — upgrade the Tracking Station to level 3, then use the "Track Unknown Objects" button in the observatory.

The Claw

The Advanced Grabbing Unit (the "Klaw") attaches to asteroids as if docking. Once grabbed, the asteroid becomes part of your craft and you can push/pull it. Tips:

- Approach slowly (asteroids are massive — 10 m/s is a collision, not a dock)
- Time warp with care — the Klaw can phase through the asteroid at high warp
- Bring extra reaction wheels — an E-class asteroid can weigh thousands of tons
- Target a Kerbin periapsis of 35 km for aerocapture with a heat shield

> **Between planets, there is silence.**
>
> The engine has cut off. The trajectory is set. For the next hundred days, your ship will coast through the void — no burns, no alarms, no ground control. Just the hum of life support and the slow rotation of stars outside the window.
>
> This is the part of spaceflight that movies skip. The waiting. The quiet. The knowledge that you are moving faster than any human ever has, and it still takes months to cross a solar system. Your kerbals are fine. They have snacks. They have each other. They have the stars.
>
> Don't time-warp through this. Not every time. Sometimes, just sit with it. The silence is part of the journey.

Player Challenges — Wave 1

Interplanetary missions that test your design and piloting skills.

- **Duna Independent:** Duna round-trip with no ISRU. Bring all fuel from Kerbin. Design a ship with 6,500+ m/s from LKO.
- **Duna Double:** Land on Duna AND Ike in the same mission. Requires lander capable of two separate descents or an SSTO lander.
- **Eve Rocks:** Land on Eve, plant a flag, and return the kerbal safely to Kerbin. The hardest stock challenge. Gilly ISRU is fair game.
- **Mohole Diver:** Land on Moho and return. Requires 8,000+ m/s from LKO. Use a nuclear transfer stage.
- **Jool-5:** Land on all 5 Jool moons (Laythe, Vall, Tylo, Bop, Pol) in one mission. Tylo is the barrier — design the mission around it.
- **Minmus Fuel Empire:** Build a fully automated Minmus mining base with miner, tanker, and orbital depot. Bonus: refuel an interplanetary ship from the depot.
- **Laythe SSTO:** Build an SSTO spaceplane that reaches Kerbin orbit, transfers to Laythe, lands, and returns — all without refueling or staging.
- **System Relay:** Deploy CommNet relay satellites to every planet and moon in the Kerbol system. Bonus: full coverage with zero dead zones.

[WAVE 2]

[The Long Ascent]

[Colonization, Life Support & Interstellar Travel]

Wave 2 — Mod List

About Wave 2

Wave 2 is the endgame. It adds life support, colonization, interstellar travel, and realism overhauls. These mods change the game fundamentally — mistakes now have permanent consequences, and the systems interact in complex ways. You should be fluent in interplanetary missions (Wave 1) and have at least one established colony or station before attempting this wave.

Criterion	Wave 0	Wave 0.5	Wave 1	Wave 2
Gameplay changed?	No	No	Adds content	Adds complexity
New parts?	No	No	Yes (2000+)	Yes (colony + interstellar)
Learning curve	None	Tools & planning	Parts, planets, graphics	Life support, colonies, FTL
Resource chains?	No	No	Fuel types (CRP/cryo)	Full (LS, colony, reactor fuel)
Failure conditions?	No (revert only)	No (revert only)	Mission failure (no revert)	Permanent (dead kerbals)
Progression goal	Stable orbit	Mun, docking, stations	Interplanetary	Interstellar, self-sustaining
Save-breaking?	No	No	Unlikely	Likely (choose wave carefully)

> Wave 2 mods are optional. Install only the ones that match your preferred challenge level. Removing a Wave 2 mod mid-save will likely break that save.

Mod List

Planetside Exploration Technologies	Colony and surface base building parts: inflatable habitats, landing pads, modular connector tubes, resource silos, and deployable surface structures. Designed for constructing permanent planetary outposts that can be expanded incrementally.

Dependencies: Community-CategoryKit, B9PartSwitch, ModuleManager

Impact: Parts, Mechanics, Gameplay

Version: Compatible with KSP 1.12.x.

Adds base-building parts optimized for surface assembly — parts ship compact and deploy on-site. Integrates with stock ISRU for fuel production. Pairs well with life support mods (USI-LS, TAC-LS) to create self-sustaining colonies. Without life support, these are purely aesthetic/RP base parts.

⚠ Conflicts: MKS (USI Modular Kolonization System) provides overlapping colonization parts and mechanics. If using MKS for the full resource-chain colonization experience, Planetside Exploration Technologies is redundant. If you only want base assembly parts without complex resource chains, use this instead of MKS.

Colonization Alternatives: Pathfinder (CKAN: "Pathfinder") and Buffalo 2 (CKAN: "Buffalo2") by Angel-125 provide a different base-building approach with inflatable modules, resource extraction, and rover parts. The Wild Blue Industries ecosystem is simpler than MKS and has broad mod support, but adds its own resource framework (WBIResources). Planetside is the recommended colonization mod for this modlist — Pathfinder and Buffalo are suggested alternatives if you prefer the WBI ecosystem or want rover-focused surface exploration.

Wave 1→2 UI Upgrades

At this stage you have hundreds of hours of flight experience. You can now replace certain Wave 0 UI mods with leaner alternatives that expect pilot proficiency:

Community Navball Docking Alignment Indicator

Impact: UI

Version: Compatible with KSP 1.12.x. Suggested replacement for DPAI in Wave 2. Uninstall DPAI if switching.

Suggested optional replacement for DPAI. By Wave 2 you no longer need a separate docking window — the navball-based indicator keeps your eyes on the instrument you're already watching during approach. No new windows, no screen clutter.

Replaces DPAI's separate alignment window with navball-integrated markers. Suggested (not required) — if you prefer DPAI's dedicated window, keep using it.

⚠ Conflicts: Docking Port Alignment Indicator — if you switch to this, uninstall DPAI. The two are mutually exclusive. Both are installed by default in Waves 0–1 so you can compare; pick one by Wave 2.

Off-World Manufacturing

By Wave 2, ISRU refueling is routine. The next step is building entire vessels from local resources — no more launching from Kerbin. These mods add the manufacturing chain to make that possible.

Extraplanetary Launchpads

Dependencies: ModuleManager

Impact: Parts, Mechanics, Gameplay

Version: Compatible with KSP 1.12.x. KIS and KAS recommended for connecting base modules and transporting resources. Infernal Robotics and KerbalStats are supported but optional.

Build rockets and bases from ore — anywhere. Adds workshops, smelters, and orbital construction docks that convert raw materials (Metal, RocketParts) into fully functional vessels. Mine ore, process it into RocketParts, and build new ships on-site without a connection to Kerbin. The definitive off-world manufacturing mod.

Adds a complete manufacturing resource chain: Ore → Metal → RocketParts. Workshops and construction docks consume RocketParts to build vessels. Part productivity depends on engineer skill. Transforms colony gameplay from "refuel here" to "the colony IS the space program." Stock ISRU and base parts already provide the mining half of the equation.

SimpleConstruction!

Dependencies: ExtraplanetaryLaunchpads, ModuleManager

Impact: Mechanics, Gameplay

Version: Compatible with KSP 1.12.x. Requires Extraplanetary Launchpads. NotSoSimpleConstruction (CKAN) adds additional EL functionality to more stock parts. Choose either SimpleConstruction (lightweight) or use EL's own parts — not both.

Gives stock parts Extraplanetary Launchpads functionality. Stock ore tanks, ISRU converters, and workshops gain the ability to produce RocketParts and build vessels — without EL's dedicated parts. If you find EL's parts ugly or don't want additional part clutter, this lets you use stock parts for the entire manufacturing chain.

Reconfigures stock parts to serve as EL workshops, smelters, and construction docks. No new part models — uses the existing stock ISRU and tank parts you already know. Requires Extraplanetary Launchpads as the underlying framework.

Interstellar Propulsion

Wave 1 gave you the tools to explore the expanded solar system (OPM, QuackPack, Minor Planets). Wave 2 interstellar propulsion makes those outer planets feel close — and lets you reach entirely new star systems (Kcalbeloh, GEP).

Far Future Technologies

Dependencies: CommunityResourcePack, B9PartSwitch, DeployableEngines, ModuleManager

Impact: Parts, Mechanics, Gameplay

Version: Compatible with KSP 1.12.x. Requires significant power infrastructure (Near Future Electrical reactors recommended). Sterling Systems (manual install) adds complementary engine parts. SystemHeat recommended for thermal management.

Advanced interstellar propulsion by Nertea: fusion torches, antimatter drives, nuclear salt-water rockets, and pulsed inertial confinement engines. Builds on the Near Future Technologies ecosystem with the same high model quality and Restock-compatible art style. The recommended interstellar propulsion mod for this modlist.

Adds endgame propulsion technologies: fusion engines (5,000–30,000s Isp), antimatter-initiated fusion, and beam-core antimatter drives (100,000+ s Isp). Requires massive power infrastructure — pairs with Near Future Electrical reactors. Engines consume advanced resources (FusionPellets, Antimatter) that must be produced or mined.

Sterling Systems

Dependencies: CommunityResourcePack, B9PartSwitch, DeployableEngines, ModuleManager

Impact: Parts, Mechanics

Version: Compatible with KSP 1.12.x. Manual install — not on CKAN. Designed to complement Far Future Technologies.

Additional advanced engine parts that complement Far Future Technologies. Adds sterling-cycle nuclear engines, hybrid thermal rockets, and niche propulsion options that fill gaps in FFT's interstellar propulsion lineup. Designed to pair with FFT for a complete near-future-to-interstellar engine progression.

Adds extra advanced engines that work with FFT's resource frameworks (FusionPellets, Antimatter). Not a standalone propulsion system — supplements FFT with additional engine variety and niche propulsion options.

Blueshift

Dependencies: ModuleManager

Impact: Parts, Mechanics, Gameplay

Version: Compatible with KSP 1.12.x. Recommends Kopernicus for interstellar destinations. Waterfall recommended for warp-drive visual effects.

FTL (faster-than-light) warp drive mod by Angel-125. Unlike instant warp drives, Blueshift requires you to build a warp-coil infrastructure — you must construct and deploy warp coils at your destination before you can FTL there. A "balanced" FTL system that makes you work for faster-than-light travel rather than handing it to you.

Adds FTL warp drives requiring destination infrastructure. You must deploy warp-coil networks before FTL travel becomes possible — no free jumps. Warp travel is gated behind progression milestones rather than handed out instantly. Uses Kopernicus for interstellar destinations.

⚠ Conflicts: Far Future Technologies provides sub-light interstellar propulsion (fusion torches, antimatter drives). Blueshift's FTL warp system is a complementary layer — use both for a complete interstellar progression: FFT engines to explore new stars, then deploy warp coils for fast travel between them.

Base Building Alternative: Kerbal Planetary Base Systems (https://forum.kerbalspaceprogram.com/topic/133606-112x-kerbal-planetary-base-systems/) is a popular alternative to Planetside Exploration Technologies. KPBS adds modular surface base parts with a distinct visual style and broad mod support. Manual install from forum thread — not on CKAN. Choose Planetside (CKAN, benjee10 style) or KPBS (manual, modular style) for your surface base needs.

EVA Construction Tools: KIS — Kerbal Inventory System (https://forum.kerbalspaceprogram.com/topic/149848-minimum-ksp-version-112-kerbal-inventory-system-kis-v129/) and KAS — Kerbal Attachment System (https://forum.kerbalspaceprogram.com/topic/23555-0211-kas-v043-struts-pipes-part-storage-containers-merged-winches-amp-more/) are the classic EVA construction pair. KIS lets kerbals carry and place parts during EVA; KAS adds winches, pipes, and struts for connecting base modules. Both are manual install from forum threads — not on CKAN. These were once essential and remain popular, though KSP 1.11+ added stock EVA construction that covers many KIS/KAS use cases.

Wave 2 Mods — How to Use Them

Planetside Exploration Technologies (Base Building)

Planetside gives you the parts to build surface bases, but it doesn't add new resource chains — it works with stock ISRU. Here's how to approach base construction:

Phase 1 — Scout: Before you land a single part, scan the target body with SCANsat (from Wave 1). Find a flat area with high ore concentration. Minmus's Flats and the Mun's craters are ideal first targets. Land a small probe to confirm the terrain is as flat as the map suggests.

Phase 2 — Power and ISRU: Land a drilling rig first. The minimum viable base: a Drill-O-Matic, a Convert-O-Tron 250, an ore tank, a fuel tank, and sufficient power (solar for inner planets, reactors for outer). Radiators are not optional — the converter will overheat and shut down within seconds without them. Test the full ISRU chain on the launch pad before flying it anywhere.

Phase 3 — Expansion: Now deploy Planetside parts. Inflatable habitats connect to the ISRU hub via modular connector tubes. Landing pads let you land subsequent supply ships precisely (they have guidance beacons). Resource silos store excess fuel and ore. Think of the base as a space station that happens to sit on the ground — the same assembly principles apply.

Phase 4 — Self-sufficiency (with life support): If you add a life support mod (USI-LS, TAC-LS, Kerbalism), your base now needs to produce supplies in addition to fuel. Greenhouses and recyclers close the resource loop. The goal: a base that produces fuel, supplies, and can even build new vessels (when paired with Extraplanetary Launchpads or similar).

> Build your first base on Minmus. The gravity is so low that landing heavy modules is trivial, and the flats have the highest ore concentration in the Kerbin system. Once you've proven the design works on Minmus, replicate it on the Mun, then Duna.

> Precision landing matters a lot when assembling a base. If your habitat module lands 5 km from your ISRU rig, you either spend hours rovering parts together or you revert and try again. Use MechJeb's Landing Guidance (Wave 1) or get very good at manual targeted landings before attempting base assembly.

Integration with Wave 0–1 Mods

By Wave 2, your modlist is fully layered. Here's how the mods work together during a typical colony mission:

- **KER** or **MechJeb** gives you delta-v and TWR readouts during base module design — heavy ISRU equipment requires powerful transfer stages
- **MechJeb's Landing Guidance** can precision-land base modules within a few hundred meters of your target
- **BetterTimeWarp** lets you warp through resource processing at custom speeds without breaking physics
- **TransferWindowPlanner** optimizes the trip from Kerbin to your colony world
- **SCANsat** provides the resource map that tells you where to land
- **Near Future reactors** (from Wave 1) power bases on outer planets where solar is useless
- **Planetside parts** provide the base structure, and stock ISRU provides the fuel chain

> This layered integration is the philosophy of the 3-wave system. Each wave builds on the previous one, and by Wave 2, your mods form a coherent whole rather than a pile of independent addons.

Community Navball Docking Alignment Indicator

By Wave 2, you should be comfortable docking without DPAI's separate window. The Navball DAI replaces it with a marker directly on the navball.

- **Green marker:** Your docking port is aligned with the target port. Approach slowly.
- **Red marker:** Misaligned. Translate (I/J/K/L) to center the marker.
- **Range ring:** A circle around the marker indicates distance. It shrinks as you get closer.
- **Prograde marker:** The standard prograde marker still shows relative velocity. Keep it centered on the DAI marker for a perfect straight-in approach.

> Switch your camera to "Chase" mode (V key) and align the view with your docking port. The navball, camera, and DAI marker all agree — approach becomes trivial.

Extraplanetary Launchpads — Building Off-World

Extraplanetary Launchpads (EL) lets you build vessels anywhere — no Kerbin required. SimpleConstruction is a lighter alternative with fewer parts.

The Build Chain

1. **Mine:** Extract MetalOre with standard drills (same as stock ore drilling).
2. **Smelt:** Convert MetalOre → Metal in a smelter part.

3. **Build:** Convert Metal → RocketParts in a workshop. RocketParts are the construction currency.
4. **Construct:** Open the EL construction window (toolbar button), select a craft file, assign a survey stake or docking port as the spawn point. The workshop consumes RocketParts over time.
5. **Launch:** The vessel materializes at the spawn point — fully fueled if you supplied tanks of fuel.

Production Rates

- A single workshop with 2 engineers produces 1 ton of RocketParts per day.
- A small probe (5 tons) takes 5 days; a large lander (50 tons) takes 50+ days.
- Multiple workshops stack. 4 workshops + 8 engineers = 4× speed.
- Time warp works during construction. Set KAC alarm for completion.

> Build a survey station (the EL survey part) near your mining base. The survey station increases build range and provides a construction waypoint. Without it, you can only build at the exact location of the workshop.

Wave 2 Guide — The Long Ascent

Strategy, Not Tutorials

Wave 2 assumes you can build, fly, dock, and interplanetary-transfer without assistance. This guide focuses on **how systems interact** and **how to plan** rather than step-by-step instructions.

Life Support Management

Understanding the Stakes

Life support mods (USI-LS, TAC-LS, Kerbalism) introduce consumable resources: supplies/food, water, oxygen, and waste management. Kerbals consume these constantly. If they run out, kerbals die — permanently or after a grace period, depending on the mod and configuration.

Supply Chain Design

1. **Short missions** (Mun, Minmus): Bring all supplies from Kerbin. A Mk1 pod with two kerbals needs 30 supplies for a Mun round trip. Trivial.
2. **Medium missions** (Duna, Eve orbit): Bring a greenhouse or recycler module. Water purifiers and CO2 scrubbers turn waste back into usable resources, reducing net consumption by 60–90%.
3. **Long missions** (Jool, Eeloo): Build self-sustaining habitats. Greenhouses running on ore + electricity produce supplies indefinitely, but require a mining outpost or regular resupply.
4. **Colonies**: Close the loop entirely. Combine ISRU, greenhouses, and waste processing. The goal is net-zero consumption.

> Always send supplies **ahead** of crewed missions. Launch an uncrewed supply depot to the destination's orbit before sending kerbals. A dead supply mission is inconvenient; dead kerbals are a mission failure.

Habitation and Homesickness

Some life support mods add a “habitation” timer — kerbals can’t stay in cramped pods indefinitely without going stir-crazy (refusing to work, then going rogue). Counter with:

- Larger habitation modules (inflatable habitats, centrifuges)
- Shared living space (more volume per kerbal)
- Colonization modules (permanent structures with high habitation multipliers)

Colonization Workflows

The Bootstrap Problem

Building a colony requires heavy infrastructure (habitats, factories, power), but launching heavy infrastructure from Kerbin is expensive. This is the core colonization challenge: **how do you make a colony that builds itself?**

The solution is staged:

1. **Survey phase** — Scan for ore, water, rare resources. Send probes with scanners.
2. **Pioneer outpost** — Minimal crew, basic drill + ISRU, a power source (solar or nuclear). This outpost produces fuel for subsequent flights but cannot build new modules.
3. **Manufacturing hub** — Extraplanetary Launchpads or MKS workshops. These consume raw materials and MetalParts/SpecializedParts to build new vessels and modules **on site**.
4. **Self-sustaining colony** — The colony produces all of: supplies (food/water/O2), fuel, construction materials, and new vessels. At this point, Kerbin is a launch pad for initial crew, not a supply line.

> MKS and USI mods are designed to work together (they share USI-LS as a common dependency). MKS provides the colonization parts and resource chains; USI-LS provides the life support mechanics. Installing one without the other is possible but you will miss the full integration.

Resource Chains — Colonization

A typical MKS-style colonization resource chain:

- **Raw materials:** Dirt, Water, Substrate, Minerals, MetallicOre, Uraninite (mined by drills)
- **Refined materials:** Chemicals, Polymers, Metals, RefinedExotics (produced by refineries)
- **Construction:** MaterialKits, SpecializedParts, Machinery (produced by workshops from refined materials)

- **Life support:** Supplies, Mulch, Fertilizer (Mulch + Fertilizer + greenhouse = more Supplies)
- **Maintenance:** Machinery slowly wears out. Workshops consume MaterialKits to produce replacement Machinery.

> This is the "Everything is a resource chain" phase of KSP. If you enjoy Factorio-style logistics puzzles, this is your moment. If it sounds exhausting, skip the colonization layer and stick with life support + interstellar travel only.

Interstellar Travel

Engine Classes

Far Future Technologies introduces propulsion technologies beyond chemical rockets:

- **Nuclear thermal (NERVA):** Heats propellant with a fission reactor. Isp 800–1,000 s. Good for interplanetary heavy lift.
- **Fusion drives:** Isp 5,000–30,000 s. Requires massive power plants (often the engine IS the power plant). Interstellar-capable.
- **Antimatter:** Isp 100,000+ s. Exotic, dangerous, requires antimatter production infrastructure.
- **Warp/Alcubierre:** Science fiction. Warp drive contracts space ahead and expands it behind. Instant or near-instant travel, requires exotic matter.

> High Isp, low thrust engines mean **burns measured in days or weeks**. A fusion drive with 0.1 m/s^2 acceleration needs hours to change velocity by 1 km/s. Plan your burns around periapsis kicks — multiple short burns at periapsis to raise apoapsis gradually. Do not attempt to circularize an interstellar trajectory with a single burn.

> **You are between stars.**
>
> Kerbol is a bright point behind you. Grannus is a dim red dot ahead. Between them: light-years of vacuum, cosmic rays, and the faint background hum of the universe. Your ship's fusion drive has been burning for years. It will burn for years more. The kerbals on board were young when they left. They will be older when they arrive.
>
> This is what interstellar travel means. Not warp drives and instant jumps — though those will come — but the slow, patient crossing of the gulf between suns. You are doing something that humans have only dreamed of. You are committing to a journey that will take longer than the entire history of aviation on Earth.
>
> The engine burns. The stars don't move. And somewhere ahead, a new sun waits.

Interstellar Destination Profiles

With Kcalbeloh and Grannus installed, you have two interstellar destinations. Each requires different preparation.

Grannus — The Red Dwarf

Grannus is a binary red dwarf companion to Kerbol. It's the easier first interstellar target:

- **Transit time:** 50–100 years with fusion engines (FFT). Faster with antimatter.
- **Planets:** Multiple terrestrial and gas giant worlds. Full biome and science support.
- **Strategy:** Grannus is a "conventional" star system — bring ISRU equipment, establish a mining base, use it as a staging point. No exotic physics required.
- **Power:** Solar panels work poorly this far from Kerbol. Bring fission reactors (Near Future Electrical) or RTGs.

> **A new sun.**
>
> For decades, Grannus was a red point of light. Now it's a disc. An alien star with alien planets, and you are the first kerbal to see them. The light is redder than Kerbol's, casting long shadows and strange colors across unfamiliar terrain.
>
> You crossed the void between stars to get here. Not in a loading screen. Not in a cutscene. In real game-time, with real physics, using engines you built and fuel you refined and courage you found somewhere between "what if" and "why not."
>
> Welcome to another solar system. It's been waiting for you.

Kcalbeloh — The Black Hole System

Kcalbeloh is a black hole with orbiting planets. Install it for a truly alien destination:

- **Wormhole option:** A wormhole near Jool connects to Kcalbeloh. You can send probes through without interstellar engines. Crewed? Nobody's tested the radiation.
- **Transit time (conventional):** 100–300 years. Antimatter drives cut this to decades.
- **Black hole effects:** Time dilation near the event horizon is visual only (no gameplay effect), but the accretion disk is spectacular. Screenshot territory.
- **System layout:** Habitable planets, gas giants, and exotic bodies orbit the singularity. Each has unique science — some require special protection (radiation shielding parts).
- **Strategy:** Send an uncrewed probe through the wormhole first. Map the system with SCANsat. Then decide whether to mount a crewed mission — the delta-v for capture in the black hole's gravity well is enormous. Plan for it.

The black hole has an accretion disk.

It is made of matter being torn apart at relativistic speeds. It glows. The light bends around the event horizon — you can see the back of the accretion disk warped above and below the black hole itself. The singularity is invisible. The effect is unmistakable. You are looking at the most violent object in the known universe, and it is beautiful.

Time moves differently here. Not in the game — the game clock ticks normally — but in your mind. You arrived. You are orbiting a black hole. Whatever you do next, you will always be a kerbal who orbited a black hole.

Take the screenshot. Everyone does. No one forgets.

Installing both Kcalbeloh AND Grannus is a significant memory commitment. If your game crashes during interstellar transit, reduce texture quality or remove one system. The guide assumes both are installed — you can still follow it with just one.

Blueshift Warp Infrastructure

Blueshift gives you FTL travel — but with a catch. You must deploy warp coils at your destination before you can jump there.

How Warp Coils Work

1. A warp coil is a deployable part (like a satellite) that you leave in orbit around a target body.
2. Once a coil is deployed and powered, any Blueshift-equipped ship can warp to it — instantly.
3. Coils consume power (significant amounts — fission reactors minimum).
4. Range is limited by coil level (basic → advanced → exotic). Upgraded coils reach further systems.

The Scout-and-Deploy Loop

1. Send a Far Future Technologies ship to the target system using sub-light engines (fusion/antimatter).
2. The scout ship carries a warp coil in its payload bay.
3. Upon arrival, deploy the coil in orbit around the destination star or a key planet.
4. The scout ship can now warp home OR subsequent Blueshift ships can warp to the coil.
5. Repeat for each new system. Your warp network grows with exploration.

> This loop is the intended progression: FFT engines for initial exploration (slow, difficult, rewarding), Blueshift for established routes (fast, convenient, earned). Without FFT, Blueshift has no way to deploy the first coil. They are designed to work together.

Coil Network Strategy

- **First coil:** Deploy at Grannus (closer, easier than Kcalbeloh). Use a fusion-powered scout with basic coil.
- **Second coil:** Kcalbeloh. Requires advanced coil and antimatter-powered scout (longer range, higher power draw).
- **Waypoint coils:** Deploy coils at gas giants between stars for emergency refueling stops.
- **Coil security:** Coils are vulnerable. If a coil loses power, ships cannot warp to it. Include redundant power (solar + fission) and a probe core for remote rebooting.

> **The stars move.**
>
> One moment you were in orbit around Kerbol. The next, the warp field collapsed and the stars rearranged themselves. Grannus is now a disc, not a dot. The journey that took your scout ship decades just happened in an instant.
>
> This is what you worked for. The warp coil you deployed. The power infrastructure you built. The years of sub-light travel to place that first beacon. It all led to this: the ability to cross light-years in heartbeats. Not because it was easy. Because you earned it.
>
> Look out the window. Those aren't your stars. That's not your sun. You are in another solar system, and you got here by bending the fabric of spacetime with technology you assembled in orbit. Welcome to the interstellar age.

Interstellar Mission Planning

1. **Power infrastructure:** Interstellar engines consume gigawatts. You need nuclear reactors, beamed power networks, or on-board fusion. Solar panels are useless beyond Duna's orbit.
2. **Radiators:** Thermal management becomes the limiting factor. Every reactor and engine produces waste heat. Without enough radiator area, your ship melts. High-temperature radiators (graphene) are more mass-efficient.
3. **Braking:** Arriving in another star system requires deceleration. Bring enough delta-v for capture — aerocapture at the target star is rarely practical at interstellar velocities.
4. **Communications:** At interstellar distances, CommNet latency becomes absurd. Plan for autonomous probe operation.

Deep Space Vessel Architecture

Interstellar ships are not just bigger rockets. They're infrastructure you assemble in orbit and live on for decades (or centuries, in game-time).

Modular Assembly in Orbit

Build your interstellar ship in Kerbin orbit using docking:
1. **Engine section:** Fusion or antimatter drive cluster with radiators. Launched first (heaviest module).
2. **Fuel section:** Cryogenic tanks for fusion fuel pellets or antimatter containment pods. Launched separately and docked.
3. **Habitation ring:** Centrifuge or inflatable habitat modules. Launched crewless, crew boards last.
4. **Lander/payload bay:** Detachable exploration vessel for the destination system.

Use the largest docking ports (Sr.) or multi-port connections for structural rigidity. A ship assembled from Jr. ports will flex apart under thrust.

Radiator Placement

Thermal management is the limiting factor for interstellar ships. Rules:
- Radiators must face away from the sun AND away from other hot components
- Place radiators perpendicular to the ship's long axis (like wings)
- Never place hab modules between engine and radiators — they'll cook
- High-temperature graphene radiators (from System Heat) are more mass-efficient but more fragile

Artificial Gravity

Kerbals on multi-decade missions need gravity or they go stir-crazy (USI-LS habitation timer). Options:
- **Centrifuge rings:** Stockalike Station Parts Redux has inflatable centrifuges. Spin them up with reaction wheels.
- **Spin gravity:** Rotate the entire ship. Needs RCS to start/stop rotation. Docking during spin is impossible — stop first.
- **Ignore it:** Send enough habitation modules that the timer exceeds mission duration. Valid strategy, more mass.

Power Hierarchy

1. **Solar:** Useless beyond Duna. Ignore for interstellar.
2. **Fission reactors** (Near Future Electrical): Reliable, long-lasting, moderate power. Good for coil power and life support.

3. **Fusion reactors:** High power, consumes fusion fuel. The engine often doubles as the power plant.
4. **Antimatter reactors:** Extreme power, extreme danger. Antimatter containment failure = ship becomes a cloud of plasma.

> Antimatter containment requires active power. If your reactor shuts down, the containment field fails, and the antimatter annihilates. Always have a backup power source isolating the antimatter pods. This is not a drill — FFT models this behavior.

Realism Overhaul — What Changes

FAR (Ferram Aerospace Research)

Stock KSP's aerodynamics are a simplified "soup" model. FAR replaces it with voxel-based aerodynamic simulation:

- Nose cones and fairings now matter enormously (stock KSP ignores them for drag)
- Aircraft stalls are real — exceed critical angle of attack and you lose lift
- Supersonic drag is realistic — Mach 1 is a barrier, not a suggestion
- Re-entry heating becomes more dangerous without proper heat shields

RealFuels / RealPlume

Fuel is no longer a generic "Liquid Fuel + Oxidizer" mix. RealFuels adds realistic propellants:

- Kerosene/LOX (RP-1): High thrust, moderate Isp, used in first stages
- Hydrogen/LOX (LH2): High Isp, low density, used in upper stages
- Hypergolics (UDMH/NTO): Storable, lower performance, used for landers and RCS
- Solid rockets: High thrust, cannot throttle, single burn only

Each fuel type has different tank volumes, boil-off rates (cryogenics evaporate), and ignition requirements.

Multi-Colony Empire Management

By Wave 2 endgame, you're managing colonies across multiple star systems. Here's how to keep it from collapsing into chaos.

Colony Specialization

Each colony should do ONE thing well:

- **Minmus Flats:** Fuel hub. Exports LFO, LH2, Monoprop. Lowest launch costs in the game.
- **Duna Midlands:** Manufacturing. Exports MaterialKits, SpecializedParts. Needs imported fuel and supplies.
- **Laythe Islands:** Science + Crew. Mobile Processing Lab in Jool orbit processes data from the entire system.
- **Vall Orbit:** Outer-system fuel depot. Services Jool missions and interstellar departures.
- **Grannus II:** Interstellar gateway. First off-world colony in another star system.

Supply Routes

- **Scheduled tankers:** Build reusable fuel tankers on fixed routes (Minmus → LKO, Vall → Jool SOI edge). Launch them on every transfer window.
- **KAC alarm chains:** Set an alarm for each transfer window. Kerbal Alarm Clock's "transfer window" alarm type does this automatically.
- **Emergency reserves:** Every colony keeps 50% more supplies than its next resupply window requires. A missed window should not mean dead kerbals.

When to Stop Expanding

- **Part count:** Each colony adds 50–200 parts. KSP's physics engine degrades above 500 total parts in physics range. Spread colonies across SOIs to avoid loading them simultaneously.
- **Diminishing returns:** Your third fuel depot doesn't add much. Focus on one excellent depot per planetary system.
- **The kraken:** Large ships docked to large stations invite physics glitches. Autostrut everything. Quicksave before docking. Accept that sometimes the universe just breaks.

Endgame — What "Winning" Looks Like

There's no win condition in KSP. But here's what a "complete" Long Ascent playthrough looks like:

1. Self-sustaining colonies at Minmus, Duna, and Laythe
2. A Jool gateway station servicing outer-planet missions
3. Warp coil network connecting Kerbol → Grannus → Kcalbeloh
4. At least one crewed interstellar round-trip with all kerbals returning alive
5. A flag planted on Tylo (because you earned it)

After that, you've beaten the modlist. Start a new save with Kerbalism's full realism settings, or try Real Solar System. The Long Ascent taught you how to fly — now fly anywhere.

System Interactions — The Full Picture

With all Wave 2 mods active, the systems interact:

1. **Life support** determines mission duration → duration determines **engine choice** (short trip or long)?
2. **RealFuels** determines engine performance → engine choice determines **delta-v budget**
3. **FAR** determines ascent efficiency → ascent efficiency determines **payload to orbit**
4. **Colonization** provides **in-situ fuel production** → reduces required launch mass from Kerbin
5. **Interstellar engines** consume **reactor fuel** (fission/fusion/antimatter) → colonization must produce reactor fuel too

> When all systems are active, Minmus becomes your most important asset. Its low gravity means cheap ISRU, its high ore concentration means efficient resource extraction, and its location at Kerbin's edge means you're already partway out of the gravity well. Build your first major colony on Minmus, not the Mun.

> **Look how far you've come.**
>
> You started on a launchpad with a command pod and a dream. Now you have colonies on three planets, a station at Jool, warp coils bridging star systems, and kerbals living on worlds you discovered. The Kerbol system is not the same place it was when you installed Wave 0.
>
> You built this. Not the game. You. Every module docked. Every fuel run completed. Every rescue mission that became a triumph. Every moment of terror when a lander tipped over and you somehow saved it. Every quiet hour in the VAB, staring at delta-v numbers, solving problems with math and creativity.
>
> KSP doesn't have an ending. But if it did, this would be it: a universe that is bigger, stranger, and more beautiful than the one you started with — because you made it that way.
>
> Thank you for taking The Long Ascent.

Player Challenges – Wave 2

Endgame challenges requiring colonies, interstellar travel, or both.

- **Duna Self-Sufficiency:** Build a Duna colony that produces all Supplies, Fuel, and RocketParts locally. No resupply from Kerbin for 5+ years.
- **Jool Gateway:** Assemble a station in Jool orbit with 200+ parts, crew capacity of 12+, and fuel reserves of 50,000+ units. It should service missions to all 5 moons.
- **Wormhole Probe:** Send an uncrewed probe through the Kcalbeloh wormhole near Jool. Map the destination system with SCANsat and transmit data back.
- **Interstellar Crewed:** Launch a crewed mission to Grannus, land on at least one planet, and return all kerbals safely to Kerbin. Use FFT engines for transit and Blueshift for return.
- **Warp Network:** Deploy Blueshift warp coils at Kerbol, Jool, Grannus, and Kcalbeloh. Any ship must be able to warp between any two coils.
- **Tylo Solo:** Land on Tylo and return using a single-stage lander — no asparagus staging, no drop tanks. Pure engineering.
- **Colony Empire:** Establish self-sustaining colonies at Minmus, Duna, Laythe, Vall, and Grannus II. Each must produce Supplies and RocketParts locally.
- **Grand Tour:** Visit every planet and moon in the Kerbol system (stock + OPM + Minor Planets + QuackPack) in a single save file. Flag on each. Interstellar destinations are bonus.

Appendix

Maneuver Reference Card

Delta-V Map — Stock Kerbol System

Approximate vacuum delta-v from low Kerbin orbit (LKO, 80 km):

Destination	**Delta-V (m/s)**
Low Kerbin Orbit (starting point)	0
Mun transfer	860
Mun capture	310
Mun landing (from low orbit)	580
Mun return to Kerbin	860
Minmus transfer	930
Minmus capture	160
Minmus landing	180
Minmus return to Kerbin	930
Duna transfer	1,100
Duna aerocapture	0–300
Duna landing (parachutes)	100–300
Duna ascent	1,400
Eve transfer	1,100
Eve capture	300
Eve landing (parachutes)	50
Eve ascent	8,000+
Jool transfer	2,000
Jool aerocapture (Laythe)	0–500
Moho transfer	2,200
Moho capture	2,400
Eeloo transfer	2,100
Eeloo capture	1,400

> These are vacuum numbers. Atmospheric launches require more — use atmospheric delta-v in the VAB. Values assume optimal transfer windows and Hohmann transfers. Add 10–20% margin for imperfect execution.

Keybinding Quick Reference

Key	Action
Space	Activate next stage
Z / X	Full throttle / Cut throttle
Shift / Ctrl	Throttle up / down
T	Toggle SAS
F	Hold SAS (temporary)
R	Toggle RCS
G	Toggle landing gear
U	Toggle lights
M	Toggle Map view
. / ,	Time warp up / down
Tab	Cycle focus in Map view
Backspace	Reset focus to current vessel
Caps Lock	Fine-control mode
Alt+L	Lock current stage
F5 / F9	Quicksave / Quickload (hold F9)
Esc	Pause menu

Acronym Glossary

- **Ap / Pe** — Apoapsis / Periapsis (highest and lowest points of orbit)
- **CoL / CoM / CoT** — Center of Lift / Mass / Thrust
- **Dv / Δv** — Delta-V (change in velocity, your "fuel budget" in m/s)
- **ISRU** — In-Situ Resource Utilization (mining and refining resources where you land)
- **KSC** — Kerbal Space Center
- **LKO** — Low Kerbin Orbit (80 km)
- **RCS** — Reaction Control System (translation thrusters for docking)
- **SAS** — Stability Augmentation System (autopilot assist)
- **SSTO** — Single Stage To Orbit
- **TWR** — Thrust-to-Weight Ratio (must be >1 to lift off)
- **VAB / SPH** — Vehicle Assembly Building / Spaceplane Hangar

Recommended Resources

- **KSP Wiki** — https://wiki.kerbalspaceprogram.com
- **Delta-V Map (community)** — https://wiki.kerbalspaceprogram.com/wiki/Delta-v_map
- **KSP Forum** — https://forum.kerbalspaceprogram.com
- **r/KerbalSpaceProgram** — https://reddit.com/r/KerbalSpaceProgram
- **CKAN Metadata** — https://github.com/KSP-CKAN/CKAN-meta
- **Scott Manley (YouTube)** — Tutorials and career playthroughs
- **Matt Lowne (YouTube)** — Build guides and mission showcases